Does Motivation Matter? A Systematic Review and Meta-Analysis of Outcomes Following Intentional Ingestion of Foreign Objects

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Abstract

Background: Intentional Ingestion of Foreign Object(s) (IIFO) is a distinct, increasingly recognised form of self-harm that poses complex management challenges. The effect of motivation on outcomes following IIFO has not been studied.

Aims: To synthesise evidence on outcomes following IIFO and determine whether patient motivation, object characteristics, or demographics influence the need for intervention, complications, or mortality.

Methods: A systematic literature search identified studies reporting intentional ingestion of non-nutritive foreign bodies in humans. Eligible studies included any design and age group reporting endoscopic, surgical, or conservative outcomes. Exclusions were accidental or substance ingestion, non-English full texts, animal studies, and reports lacking motivation, object, or outcome data. Screening was performed by a single reviewer with 10% double-checked. Data were extracted at both study and case level. Meta-analysis of proportions summarised outcomes, and associations were explored using χ^2 testing and univariate meta-regression.

Results: Seventy-one case reports ($n = 71$) and three case series ($n = 90$) were included. Pooled outcome rates were: endoscopy 50%, surgery 30%, conservative management 20%, complications 30%, and mortality < 1%. Case series populations were uniformly male and detained, with protest motivations (≈ 80%) and sharp objects (≈ 90%) predominant. Protest motivation significantly reduced surgical intervention (OR = 0.98, 95% CI 0.98–0.98, $p = 0.003$). Intent-to-harm increased surgical risk six-fold (OR = 5.68, 95% CI 1.43–22.64, $p = 0.020$). Adults aged 40–64 had lower surgical rates but higher mortality, often with psychiatric comorbidity.

Conclusion: Motivation significantly influences IIFO outcomes. Protest ingestion may be managed conservatively, whereas intent-to-harm and psychiatric comorbidity indicate higher surgical risk. Improved prospective reporting is needed.

Keywords: Self-harm, Foreign Body, Foreign Object, Ingestion, Motivation

1 Introduction

1.1 Rationale

As of May 2024, over 100 million individuals were forcibly displaced worldwide [124]. Refugees and asylum seekers often endure extreme hardship—including violence, trauma, and detention—leading to elevated rates of mental health disorders [12, 16, 22, 88, 110, 123].

Among the most alarming manifestations is self-harm, which is up to 216 times more common in offshore detention settings than in the general population [47, 53, 126]. Methods vary and include cutting, poisoning, hanging, self-immolation, and intentional ingestion of foreign objects (IIFO) [2, 36, 53].

IIFO – defined as the non-accidental true ingestion of non-nutritive items – is a serious clinical issue, with 10–20% of cases requiring endoscopy and up to 1% needing surgery [18, 27, 58]. In displaced populations, delayed access to care increases risks [127].

Rates of intentional ingestion of foreign objects (IIFO) are rising globally and vary across populations. In the United States, incidence nearly doubled between 1997 and 2017, from 3 to 5.3 cases per 100,000 people [55]. Among adults from lower socioeconomic backgrounds, up to 92% of ingestions are intentional [55, 92]. In the UK, a forensic mental health study reported 133 IIFO incidents in a single year, involving just 27 patients – equating to one episode every 2.7 days [119].

Complications vary widely. While most involve perforation, gastro-intestinal bleeding, obstruction and mucosal erosion [58], rare incidents of aortoesophageal perforation, fistula and haemothorax [56], cardiac injury [121], liver [134] and pancreatic injury [28] are reported in the literature.

While techniques for foreign body removal have advanced—from early gastrotomy to modern endoscopy—clinical outcomes depend on multiple factors including object characteristics, patient co-morbidities, and timeliness of intervention [17, 32, 58, 59, 74, 85, 100, 102].

IIFO motivations vary widely. In detention, it may be a form of protest or communication [78, 97]; in psychiatric contexts, it may stem from conditions like psychosis, personality disorders, pica, or malingering [1, 4, 46, 78, 93, 96, 119]. In borderline personality disorder, it may function as emotional regulation rather than a suicide attempt [46]. Rare cases of treatment-resistant recurrent IIFO have prompted palliative approaches to care [60].

Despite this, little research explores how these differing motivations affect clinical management and outcomes across different populations [25, 50, 52]. Understanding motivation is crucial, as it may influence decisions around intervention [7, 14].

This review aims to examine how motivation shapes IIFO outcomes – specifically rates of endoscopic and surgical intervention, conservative management, complications, and mortality – to better inform treatment in this vulnerable patient group.

1.2 Objectives

The primary object of this systematic review was to quantify the rates of endoscopy, surgery, death, complication and conservative management following intentional ingestion of foreign objects in human populations. The review sought to examine how individual factors such as demographic/population characteristics, object characteristics and motivations for ingestion influence the likelihood of these outcomes.

2 Methods

Ethical approval was not required as all analysis was based on published data. Eligibility criteria were structured using the PICOS (Population, Intervention, Comparator, Outcome, Studies) framework.

This review was conducted in accordance with PRISMA Preferred Reporting for Systematic Reviews [90]. The protocol can be found on Github.

2.1 Eligibility Criteria

A full list of eligibility criteria is shown in Table 1. This is reproduced in a larger format for clarity in Appendix A.1 in Table A1. A full list of exclusion criteria is available in Table A2 in Appendix A.1 and in the PRISMA diagram shown in Figure 1.

2.2 Information Sources

Relevant articles were identified through a systematic search of PubMed, Web of Science, Embase, Scopus, PsycINFO, Cochrane Central Register of Controlled Trials (CENTRAL) and Google Scholar during January 2025, with the assistance of a librarian. Included articles then had their bibliography's searched by the primary author (JGE) on 14th May 2025 to identify any potential additional literature not uncovered in

Category	Details
Population	Any human; any age group.
Interventions or exposures	Non-accidental ingestion of a true foreign body (non-nutritive items).
Comparators / Control group	Demographics: Gender, age, detained person, psychiatric inpatient, displaced person, under influence of alcohol, psychiatric history, severely disabled, previous ingestion. Motivation: Intent to harm, psychiatric, psychosocial, protest, other. Object characteristics: Button battery, magnet, long (>5 cm), large diameter (>2.5 cm), multiple, blunt objects, sharp-pointed objects.
Outcomes of interest	Endoscopic intervention, surgical intervention, conservative management, complication rates, mortality.
Setting	Any setting.
Study designs	Any design.

Table 1 Inclusion criteria structured using the PICOS framework.

the primary search. The search was conducted using keywords and MeSH terms based on the concepts underpinning this review. The search queries, keywords and MeSH terms used can be found in Appendix A.2.

All identified articles were collated and duplicate articles were identified and removed. Remaining articles underwent independent title and abstract screening conducted by the first author (JGE). A randomly selected 10% sample of these articles underwent independent screening by a second reviewer (MS). Any discrepancies identified between these two reviewers were resolved by a third reviewer (GC). Inter-reviewer agreement was calculated using Cohen's kappa [79]. Remaining articles proceeded to full text review, where the same independent screening process was repeated on full text articles.

2.3 Data Extraction

Data were extracted by a single reviewer (JGE) into *Microsoft Excel* [82] and processed in *Python* [98] using *Pandas* [117]. This process is outlined in Appendix A.5.

Data was first extracted from case reports. Predictors were grouped into five subgroups: gender; age group; demographic characteristics, motivation; object characteristics. Ages were grouped into age groups based on clinical relevance. Outcome data were extracted for rates of endoscopy, surgery, conservative management, mortality, and complications. All outcomes were binary and coded per event, rather than per individual. Predictor variables and outcome variables were not mutually exclusive, nor were outcomes. For example, patients, or ingesters – hereafter referred to as the latter – could have multiple outcomes (e.g. endoscopy and surgery) and multiple predictors from each group (e.g. intent-to-harm and psychiatric, and detained and displaced person).

After case report data extraction, data was collapsed and aggregated to form a series. This data was used as a template for case series data extraction to homogenise data and reduce heterogeneity.

Full definitions of all variables (predictors and outcomes) are provided in Appendix A.5. The full dataset of extracted case-level and series-level data (including bias assesments), is available on Github and online.

2.4 Risk of Bias Assessment

Risk of bias was assessed manually for all included studies by a single reviewer (JGE), using the *Joanna Briggs Institute (JBI) Critical Appraisal Checklists for Case Reports and Case Series* [86]. Studies were first classified as either case reports or case series based on the level of granularity in the data. Each study was then evaluated using the corresponding JBI tool. A novel computational risk of bias filter was then applied in *Pandas* [117]. That process is outlined in Appendix A.6.

2.5 Synthesis Methods

For case-level associations between binary predictors and outcomes were assessed using χ^2 Test and Fisher's exact test, reporting odds ratios (ORs), 95% confidence intervals, and p-values. Where appropriate, further χ^2 tests were also used to evaluate differences in outcome proportions between groups [79].

Univariate logistic regression was considered but ultimately not used, as the primary aim was to explore associations without assuming a specific functional relationship between predictors and outcomes. Given the binary nature of most predictors and the categorical outcomes, χ^2 tests offered a more transparent and assumption-light approach. This choice also avoided complications from sparse data and convergence issues that can arise with logistic models in small exploratory datasets.

Multivariate logistic regression was not performed due to the limited sample size, high co-linearity between predictors (e.g. overlapping motivations), and the exploratory nature of the analysis.

For series-level data, univariate meta-regression will be conducted to assess associations between binary series-level predictors and pooled outcome proportions, where sufficient data are available. Each predictor will be entered separately to account for incomplete reporting across studies and to avoid over-fitting. Effect estimates were reported as odds ratios (ORs) with Restricted Maximum Likelihood (REML) estimation [113].

Initially, a meta-analysis of outcome proportions was conducted using a random-effects model to estimate pooled outcome rates across included studies. The random-effects approach was chosen due to the anticipated heterogeneity in study populations, motivations, and object types to analyse the effect of pooling case reports.

REML estimation was used to compute between-study variance (τ^2), while Hartung–Knapp (HK) adjustments were applied to produce more accurate confidence intervals, particularly in the presence of small sample sizes.

Heterogeneity was quantified using the I^2 statistic, which describes the percentage of total variation across studies that is due to true between-study differences rather than chance. Following the Cochrane Handbook's guidelines, I^2 values were interpreted as follows: 0–40% may not be important; 30–60% may indicate moderate heterogeneity; 50–90% may represent substantial heterogeneity; and 75–100% may reflect considerable heterogeneity [54]. These thresholds are intended as general guidance rather than strict rules and were interpreted in the context of the number of studies, consistency of effect sizes, and confidence interval overlap.

First, meta-analysis was undertaken on case series alone, and then on case series with pooled case report data. This method was given in anticipation of the case reports introducing heterogeneity into the case series meta-analysis.

Due to the inclusion of primarily case reports and small case series, formal assessment of reporting bias (e.g., via funnel plots or statistical tests for asymmetry) was not feasible.

Confidence in the body of evidence was not formally graded but was considered low to very low due to reliance on uncontrolled observational designs, small sample sizes, and incomplete reporting.

3 Results

3.1 Study Selection

Details on the screening and selection process are demonstrated in Figure 1. Detail on independent inter-reviewer agreement and third author review can be found in Appendix A.4. Significantly, 135 reports were excluded for failing to explicitly state whether the ingestion report was intentional. Another 91 were excluded for not reporting motivation.

3.2 Risk of Bias

Case Reports

75 cases from 67 studies [3–6, 8–10, 13–15, 19–21, 23, 24, 26, 29–31, 33–35, 38, 40, 41, 43, 45, 48, 49, 51, 61–64, 66, 68–70, 72, 75–78, 80, 84, 87, 89, 95, 99, 101, 103, 104, 106, 108, 109, 111, 112, 114–116, 118, 120, 128, 130–133] were evaluated using the *JBI Checklist for Case Reports* [86]. 3 cases were excluded. Cases were excluded at this stage if they failed to describe the following domains: patient history and timeline (1 case) [72], current patient condition (2 cases) [72], interventions and treatments (1 case) [106], patient post-intervention condition (2 cases) [72], harms (2 cases) [72], and takeaway lessons (2 cases) [72]. The excluded cases came from the following studies: [72, 106]. Of the remaining 71 cases, all reported interventions and treatments (71 cases, 100%) [3–6, 8–10, 13–15, 19–21, 23, 24, 26, 29–31, 33–35, 38, 40, 41, 43, 45, 48, 49, 51, 61–64, 66, 68–70, 75–78, 80, 84, 87, 89, 95, 99, 101, 103, 104, 108, 109, 111, 112, 114–116, 118, 128, 130–133]. Most clearly described patient history and timeline (70 cases, 99%) [3–6, 8–10, 13–15, 19–21, 23, 24, 26, 29, 31, 33–35, 38, 40, 41, 43, 45, 48, 49, 51, 61–64, 66, 68–70, 75–78, 80, 84, 87, 89, 95, 99, 101, 103, 104, 108, 109, 111, 112, 114–116, 118, 128, 130–133], patient post-intervention condition (69 cases, 97%) [3–6, 8–10, 13–15, 19–21, 23, 24, 26, 29–31, 33–35, 38, 40, 41, 43, 45, 48, 49, 51, 61–64, 66, 68, 69, 75, 77, 78, 80, 84, 87, 89, 95, 99, 101, 103, 104, 108, 109, 111, 112, 114–116, 118, 128, 130–133], takeaway lessons (69 cases, 97%) [3–6, 8–10, 13–15, 19–21, 23, 24, 26, 29, 31, 34, 35, 38, 40, 41, 43, 45, 48, 49, 51, 61–64, 66, 68–70, 75–78, 80, 84, 87, 89, 95, 99, 101, 103, 104, 108, 109, 111, 112, 114–116, 118, 128, 130–133], patient demographic (68 cases, 96%) [3–6, 8–10, 13–15, 19, 21, 23, 24, 26, 29–31, 33–35, 38, 40, 41, 43, 45, 48, 49, 51, 61–64, 66, 68–70, 75–77, 80, 84, 87, 89, 95, 99, 101, 103, 104, 108, 109, 111, 112, 114–116, 118, 128, 130, 132, 133],

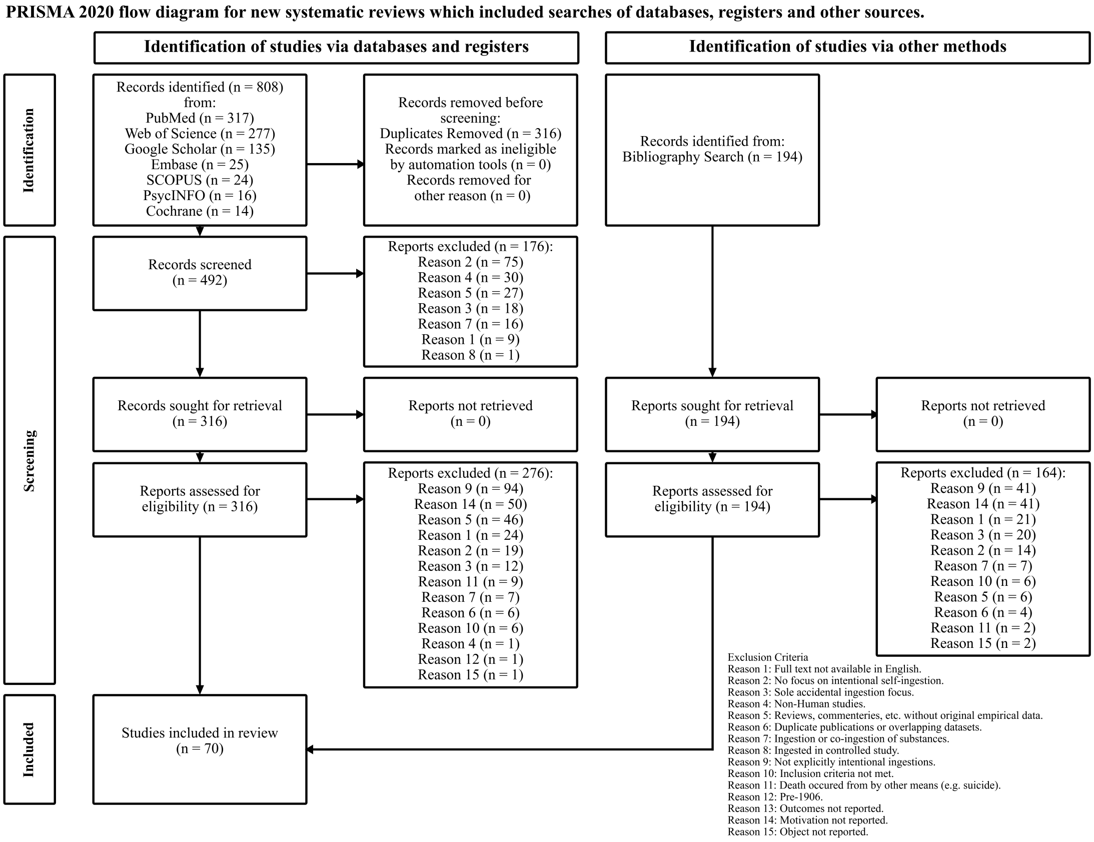

Fig. 1 PRISMA flow diagram summarising the study selection process.

and current patient condition (68 cases, 96%) [3–6, 8–10, 13–15, 19–21, 23, 24, 26, 29–31, 33–35, 40, 41, 43, 45, 48, 49, 51, 61–64, 66, 68–70, 75–78, 80, 84, 87, 89, 95, 99, 101, 103, 104, 108, 109, 111, 112, 114–116, 118, 128, 130–133]. Reporting was also strong for harms (38 cases, 93%) [8, 10, 19–21, 23, 24, 26, 29, 31, 34, 40, 43, 45, 63, 64, 68, 69, 76, 77, 84, 87, 89, 109, 111, 112, 114, 116, 118, 128, 130, 131, 133], and diagnostic assessments (65 cases, 92%) [3–6, 8–10, 13–15, 19–21, 23, 24, 26, 29, 31, 33–35, 40, 41, 43, 45, 48, 51, 61–64, 66, 68–70, 75–78, 80, 84, 87, 89, 95, 99, 101, 103, 104, 108, 109, 111, 112, 114–116, 118, 128, 130–133].

Case Series

Separately, 3 studies [39, 65, 73] were evaluated using the *JBI Checklist for Case Series* [86]. Reporting quality was generally high across all JBI domains. All included case series fully reported clear inclusion criteria, standard condition measurements, valid patient identification methods, complete inclusion, clear demographic information, clear clinical information, clear outcome and follow-up, and appropriate statistical analysis [39, 65, 73]. However, fewer studies (2) reported consecutive inclusion, and clear site demographic information [39, 65].

3.3 Study Characteristics

Case Reports

A total of 71 cases were reported 33 countries [3–6, 8–10, 13–15, 19–21, 23, 24, 26, 29–31, 33–35, 38, 40, 41, 43, 45, 48, 49, 51, 61–64, 66, 68–70, 75–78, 80, 84, 87, 89, 95, 99, 101, 103, 104, 108, 109, 111, 112, 114–116, 118, 128, 130–133].

The top three countries represented were the United States of America ($n = 12$), India ($n = 7$), and the United Kingdom ($n = 7$). The median number of case reports per country was 1 (IQR = 1.0, range 1-12).

Cases were present from a wide age range (7 to 100 years) with a median age of 28 year (IQR = 18). The majority of cases were reported in males (60% vs 39%), with 1 case of unknown gender.

Half of cases had a psychiatric history and over a quarter (26%) had ingested previously. 17% were detained, 6% were psychiatric inpatients, 10% had a severe disability and 3% ($n = 2$) were displaced persons.

Psychiatric motivation was reported in nearly half of cases (48%), with intent-to-harm and psychosocial reported in 29% and 23% of cases respectively. Protest was reported in 11% of cases and other in 12%.

Table 2 Case-level summary statistics.

Variable	Count	Percentage	References
Age			
Max	100		[75]
Mean	30		
Median	28		
IQR	18		
Min	7		[76, 89]
Age Group			
Adult (25–39)	25	35%	[6, 9, 13, 14, 20, 23, 29, 33, 34, 41, 43, 45, 49, 62, 70, 77, 84, 99, 101, 103, 104, 111, 118, 131, 133]
Young Adult (18–24)	17	24%	[3, 10, 15, 24, 35, 63, 64, 68, 77, 78, 80, 95, 108, 132]
Child/Adolescent (<18)	13	18%	[5, 8, 31, 38, 48, 76, 87, 89, 112, 114, 130]
Middle-Aged (40–64)	11	16%	[4, 26, 30, 40, 51, 61, 69, 109, 115, 116, 128]
Older Adult (65+)	3	4%	[19, 66, 75]
Gender			
Male	43	61%	[3, 4, 6, 9, 10, 13, 15, 20, 21, 26, 30, 35, 40, 41, 43, 45, 49, 61, 62, 68–70, 76–78, 80, 84, 99, 101, 108, 111, 112, 114–116, 118, 128, 132]
Female	27	38%	[5, 8, 14, 19, 23, 24, 31, 33, 34, 38, 48, 51, 63, 64, 66, 75, 87, 89, 95, 103, 104, 109, 130, 131, 133]
Unknown	1	1%	[29]
Demographic			
Psychiatric History	36	51%	[5, 6, 8, 13–15, 19, 29, 30, 33, 38, 41, 43, 49, 51, 61–63, 66, 68–70, 76, 80, 84, 89, 95, 103, 104, 109, 111, 112, 115, 133]
Prior Ingestion	19	27%	[6, 13, 21, 24, 29, 35, 38, 40, 49, 61, 62, 76, 103, 112, 115, 116, 133]
Detained	12	17%	[6, 9, 13, 77, 78, 99, 111, 118]
Severe Disability	7	10%	[15, 66, 76, 89, 95, 115, 133]
Psychiatric Inpatient	4	6%	[29, 38, 115]
Alcohol Influence	3	4%	[20, 35, 116]
Displaced Person	2	3%	[3, 45]
Motivation			
Psychiatric	34	48%	[4, 6, 8, 13–15, 23, 24, 38, 40, 41, 49, 51, 61–64, 66, 68–70, 75, 76, 84, 89, 103, 104, 109, 111, 112, 115, 132]
Intent-to-Harm	21	30%	[4–6, 10, 29, 30, 33–35, 43, 75, 77, 78, 80, 103, 111, 112]
Psychosocial	16	22%	[3, 20, 24, 31, 48, 51, 68, 75, 87, 99, 101, 108, 114, 116, 130, 131]
Other	9	13%	[8, 9, 40, 49, 95, 103, 118, 128, 133]
Protest	8	11%	[26, 45, 77, 78]
Object			
Large (>2.5cm) Diameter	51	72%	[3–6, 9, 10, 13, 15, 21, 23, 29–31, 33–35, 38, 40, 45, 49, 61, 62, 64, 66, 68–70, 77, 78, 80, 84, 87, 89, 95, 99, 101, 103, 109, 112, 115, 116, 118, 131, 133]
Multiple	44	62%	[8, 13–15, 19, 24, 26, 29–31, 40, 41, 43, 48, 49, 51, 61–64, 68–70, 75–77, 80, 84, 87, 89, 108, 109, 111, 112, 114–116, 128, 130, 132]
Sharp	34	48%	[5, 6, 13, 14, 20, 23, 24, 30, 35, 38, 40, 41, 43, 49, 51, 61, 62, 64, 68, 70, 77, 78, 80, 84, 104, 108, 115, 132]
Long (>5cm)	32	45%	[4, 5, 9, 10, 15, 23, 30, 33–35, 38, 40, 43, 45, 62, 64, 66, 68, 70, 80, 84, 89, 99, 103, 109, 115, 116, 118, 132, 133]
Magnet	9	13%	[8, 26, 31, 76, 87, 89, 112, 114, 130]
Button Battery	2	3%	[21, 26]
Outcome			
Complication	47	66%	
Surgery	43	61%	
Endoscopy	31	44%	
Conservative	7	10%	
Death	2	3%	

Most ingestions involved large (> 2.5cm) diameter (72%), sharp (62%) objects and multiple object ingestion (62%). Almost 50% of cases (48%) involved sharp and long (> 5cm, 45%) object ingestion. Fewer ingestions involved magnets (12%) and button batteries (2%)

Complication rates were high (66%), as were rates of endoscopy and surgery (61% and 44% respectively). Cases were only managed conservatively 10% of the time. The mortality rate in case reports was 2.8%.

A table of case-level characteristics is shown in Table 2.

Case Series

3 studies were case series, yielding 90 cases [39, 65, 73]. Case series were present from the United States of America [65] ($n = 19$), South Korea [73] ($n = 52$) and Tunisia [39] ($n = 19$).

Values reported herein are mean averages across all case series. Unreported variables are treated as 0.

All cases were male, aged 17–50 years and detained 33% had a psychiatric history (range 82-95%) - no psychiatric inpatients.

Demographic predictors were poorly recorded. Previous ingestion rates were only reported in one series at 11%. There was no severe disability (not reporting in two series). Data on displaced person and alcohol influence were not reported at all.

Motivations were predominantly protest (78%, range 16-97%); psychiatric in 13% (63.2% in one series, 0% in the other two); intent-to-harm in 7% (range 0-21%). There were no reports of other motivation or psychosocial motivation.

One series only report sharp object ingestion and no other object characteristics [65]. In the other two series, overall ingestion involed a sharp object in most cases (76%, range 64-100%). Long objects were ingested 37% of cases (36%, range 0-67%), multiple objects were ingested in (27.8%, range 0-67%)

In terms of outcomes, complications were only reported in one series [73], data was absent from the other two. Endoscopy occuring in 52% (range 5-89%), although it was unreported in one series [65]. Surgery occured following 17% of ingestions (range 12-26%), conservative management following 32% (0% in one case and above 70% in the other two).

A full list of grouped series-level characteristics and outcomes is available in Table 4. A map of all cases from case reports and case series is shown graphically in Figure 2 and in tabular format in Table 3.

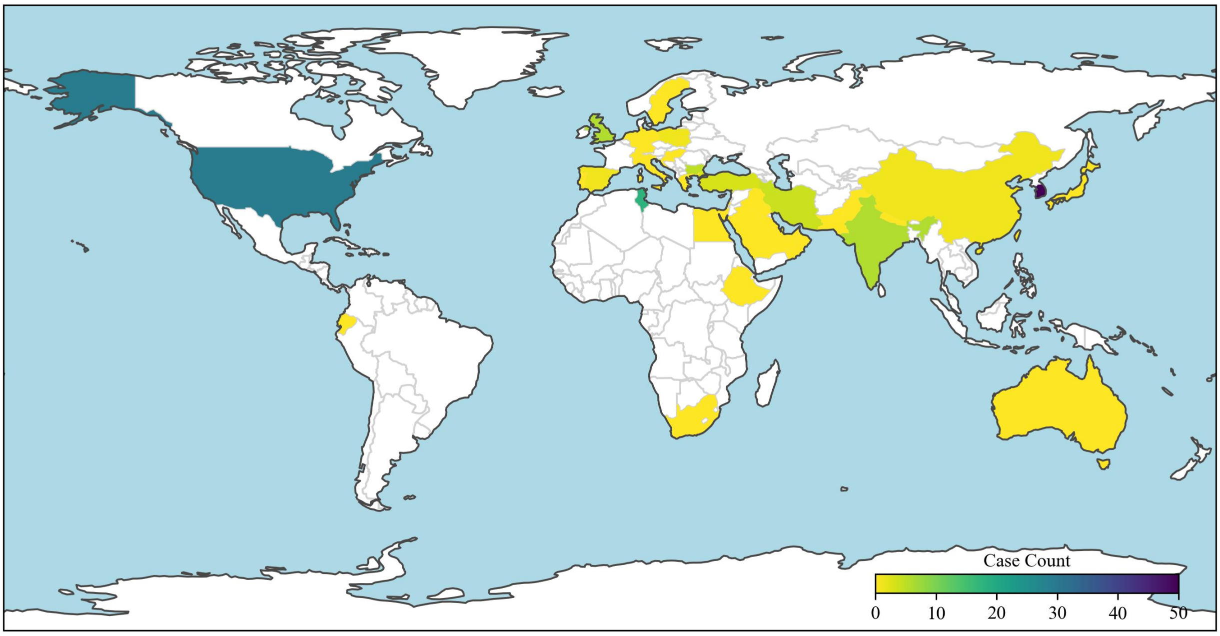

Fig. 2 Heat map of all cases per country.

3.4 Synthesis

3.4.1 Univariate Association Testing

In case reports, middle-age (40–64 years) was associated with significantly reduced odds of surgery (OR = 0.19, 95% CI: 0.04–0.78, $p = 0.020$). There were non-significant trends towards increased of endoscopy (OR 2.62, 95% CI: 0.69-9.95, $p = 0.192$) and conservative management (OR 2.44, 95% CI: 0.41-14.57, $p = 0.295$)

Table 3 Case counts per country.

Country	Case Count	%	References
South Korea	52	32.3%	[73]
United States of America	31	19.3%	[6, 14, 26, 43, 49, 51, 61, 65, 66, 69, 76, 111, 114]
Tunisia	19	11.8%	[39]
United Kingdom	7	4.3%	[19, 21, 31, 34, 45, 99]
India	7	4.3%	[23, 24, 63, 64, 70, 84, 128]
Bulgaria	6	3.7%	[77, 78]
Iran	5	3.1%	[38, 40, 41]
Turkey	4	2.5%	[3, 15, 112, 133]
Spain	2	1.2%	[29, 30]
Poland	2	1.2%	[68, 131]
China	2	1.2%	[62, 75]
Pakistan	1	0.6%	[132]
Switzerland	1	0.6%	[130]
Taiwan	1	0.6%	[33]
South Africa	1	0.6%	[108]
Saudi Arabia	1	0.6%	[109]
Qatar	1	0.6%	[10]
Portugal	1	0.6%	[95]
Sweden	1	0.6%	[87]
Australia	1	0.6%	[13]
Oman	1	0.6%	[5]
Netherlands	1	0.6%	[20]
United Arab Emirates	1	0.6%	[8]
Japan	1	0.6%	[89]
Italy	1	0.6%	[101]
Israel	1	0.6%	[48]
Iraq	1	0.6%	[4]
Hungary	1	0.6%	[35]
Greece	1	0.6%	[103]
Germany	1	0.6%	[115]
Ethiopia	1	0.6%	[80]
Egypt	1	0.6%	[9]
Ecuador	1	0.6%	[104]
Croatia	1	0.6%	[118]
Nepal	1	0.6%	[116]

Note: % rounded to one decimal place.

Further analysis of this subgroup ($n = 11$ vs $n = 60$) revealed a significantly higher proportion of males and individuals ingesting multiple objects (+25% and +8%, respectively). There was a non-significant tendency toward higher rates of psychiatric motivation (+13.5%) and history of prior ingestion (+9.3%). Conversely, fewer individuals in this age group were detained, and intent-to-harm motivation appeared less common.

Two deaths occurred in this age group, outside of a detention context: one from metal ingestion in the context of pica in schizophrenia [69] and another from drug-induced acuphagia [40].

There were no other observed statistical relationships in any of the age groups.

Results of univariate association testing are shown in Table 5.

Intent-to-harm motivation was associated with increased odds of surgery (OR = 5.68, 95% CI: 1.43–22.64, $p = 0.020$). This subgroup ($n = 21$ vs $n = 50$) included significantly more young adults (+20%), males (+16%), and individuals ingesting large-diameter objects (+6%) and sharp objects (+4%). A significant reduction in psychosocial co-motivation was observed in this group (–25%).

The presence of other motivation was associated with significantly reduced odds of surgery (OR = 0.15, 95% CI: 0.03–0.77, $p = 0.024$). There were non significant increases in the odds of conservative management (OR 1.17, 95% CI: 0.12, 10.99, $p = 1.000$), endoscopy (OR 2.96, 95% CI: 0.68-12.95, $p = 0.165$), death (OR 7.62, 95% CI: 0.43-134.24, $p = 0.239$)

Table 4 Grouped series-level summary.

Variable	**Average**	**Karp et al. [65]**	**Lee et al. [73]**	**Elghali et al. [39]**
Total Cases	90	19	52	19
Age				
Mean	24	24	—	24
Min	17	17	25	19
Median	35	—	35	—
Max	50	40	50	27
Gender				
Male	90 (100%)	19 (100%)	52 (100%)	19 (100%)
Female	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Unknown	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Demographic				
Detained	90 (100%)	19 (100%)	52 (100%)	19 (100%)
Psychiatric History	30 (33%)	18 (95%)	9 (18%)	2 (12%)
Prior Ingestion	2 (2%)	—	—	2 (10%)
Psychiatric Inpatient	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Displaced Person	0 (0%)	—	—	—
Alcohol Influence	0 (0%)	—	—	—
Severe Disability	0 (0%)	—	0 (0%)	—
Motivation				
Protest	70 (78%)	3 (16%)	50 (97%)	17 (90%)
Psychiatric	12 (13%)	12 (63%)	0 (0%)	0 (0%)
Intent-to-Harm	6 (7%)	4 (21%)	0 (0%)	2 (10%)
Psychosocial	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Other	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Object				
Sharp	68 (76%)	19 (100%)	33 (64%)	16 (84%)
Long (>5cm)	32 (36%)	—	32 (62%)	0 (0%)
Multiple	25 (28%)	—	24 (46%)	1 (5%)
Button Battery	0 (0%)	—	0 (0%)	0 (0%)
Magnet	0 (0%)	—	0 (0%)	0 (0%)
Large Diameter (>2.5cm)	0 (0%)	—	—	—
Outcome				
Endoscopy	47 (52%)	—	46 (88%)	1 (5%)
Conservative	29 (32%)	14 (74%)	0 (0%)	15 (79%)
Surgery	15 (17%)	5 (26%)	6 (12%)	4 (21%)
Complication	6 (7%)	—	6 (12%)	—
Death	1 (1%)	0 (0%)	0 (0%)	1 (5%)

Key: n (%). Average = predictor and outcome mean average rates across series.

Table 5 Univariate associations between predictors and outcomes.

Variable	Conservative	Endoscopy	Surgery	Death	Complication
Age Group					
Child/Adolescent (<18)	1.93 [0.33, 11.24] (p=0.604)	0.32 [0.08, 1.29] (p=0.178)	2.53 [0.63, 10.15] (p=0.307)	—	1.89 [0.47, 7.65] (p=0.521)
Young Adult (18–24)	1.31 [0.23, 7.44] (p=0.670)	0.31 [0.09, 1.06] (p=0.101)	2.60 [0.75, 9.01] (p=0.210)	—	1.30 [0.40, 4.25] (p=0.885)
Adult (25–39)	0.28 [0.03, 2.45] (p=0.409)	2.17 [0.81, 5.85] (p=0.195)	0.96 [0.36, 2.61] (p=1.000)	—	0.86 [0.31, 2.39] (p=0.979)
Middle-Aged (40–64)	2.44 [0.41, 14.57] (p=0.295)	2.62 [0.69, 9.95] (p=0.192)	**0.19 [0.04, 0.78] (p=0.020)***	—	0.56 [0.15, 2.05] (p=0.490)
Older Adult (65+)	—	0.63 [0.05, 7.32] (p=1.000)	1.32 [0.11, 15.25] (p=1.000)	—	1.02 [0.09, 11.88] (p=1.000)
Gender					
Male	0.22 [0.04, 1.25] (p=0.104)	1.34 [0.51, 3.53] (p=0.722)	0.99 [0.37, 2.62] (p=1.000)	—	1.49 [0.55, 4.06] (p=0.595)
Female	4.77 [0.86, 26.63] (p=0.097)	0.82 [0.31, 2.18] (p=0.887)	0.92 [0.34, 2.44] (p=1.000)	—	0.79 [0.29, 2.17] (p=0.847)
Demographic					
Detained	—	0.96 [0.27, 3.42] (p=1.000)	0.89 [0.25, 3.18] (p=1.000)	—	0.64 [0.18, 2.32] (p=0.515)
Psychiatric Inpatient	—	0.43 [0.04, 4.40] (p=0.630)	2.32 [0.23, 23.75] (p=0.630)	—	0.15 [0.01, 1.54] (p=0.108)
Displaced Person	—	—	0.67 [0.03, 12.84] (p=1.000)	—	0.50 [0.03, 9.77] (p=1.000)
Alcohol Influence	—	0.83 [0.07, 10.20] (p=1.000)	1.05 [0.09, 12.88] (p=1.000)	—	—
Psychiatric History	0.92 [0.19, 4.51] (p=1.000)	0.74 [0.27, 2.06] (p=0.749)	1.45 [0.52, 4.07] (p=0.657)	0.69 [0.04, 11.51] (p=1.000)	0.74 [0.25, 2.17] (p=0.780)
Severe Disability	—	4.02 [0.72, 22.47] (p=0.120)	0.83 [0.17, 4.04] (p=1.000)	—	1.35 [0.24, 7.54] (p=1.000)
Prior Ingestion	1.62 [0.29, 9.05] (p=0.669)	0.78 [0.24, 2.50] (p=0.902)	0.78 [0.24, 2.51] (p=0.911)	1.56 [0.09, 26.48] (p=1.000)	0.36 [0.10, 1.29] (p=0.203)
Motivation					
Protest	—	0.39 [0.07, 2.09] (p=0.447)	4.81 [0.55, 41.92] (p=0.239)	—	4.12 [0.47, 35.95] (p=0.247)
Psychiatric	6.86 [0.78, 60.48] (p=0.105)	1.37 [0.52, 3.61] (p=0.699)	0.49 [0.18, 1.33] (p=0.248)	—	0.70 [0.25, 1.94] (p=0.670)
Psychosocial	1.29 [0.22, 7.37] (p=1.000)	0.99 [0.32, 3.08] (p=1.000)	0.56 [0.18, 1.75] (p=0.482)	—	0.86 [0.27, 2.76] (p=1.000)
Intent-to-Harm	—	0.47 [0.15, 1.48] (p=0.307)	**5.68 [1.43, 22.64] (p=0.020)***	—	0.88 [0.29, 2.67] (p=1.000)
Other	1.17 [0.12, 10.99] (p=1.000)	2.96 [0.68, 12.95] (p=0.165)	**0.15 [0.03, 0.77] (p=0.024)***	7.62 [0.43, 134.24] (p=0.239)	0.60 [0.14, 2.46] (p=0.475)
Object					
Sharp	1.51 [0.31, 7.30] (p=0.703)	**0.32 [0.12, 0.85] (p=0.037)***	2.27 [0.85, 6.06] (p=0.157)	1.09 [0.07, 18.15] (p=1.000)	1.13 [0.42, 3.04] (p=1.000)
Long (>5cm)	0.44 [0.08, 2.44] (p=0.442)	0.76 [0.29, 1.97] (p=0.746)	2.56 [0.94, 6.94] (p=0.106)	1.19 [0.07, 19.88] (p=1.000)	1.96 [0.70, 5.49] (p=0.304)
Multiple	4.11 [0.47, 36.14] (p=0.240)	0.46 [0.17, 1.21] (p=0.181)	1.09 [0.41, 2.91] (p=1.000)	—	2.79 [1.01, 7.71] (p=0.081)
Button Battery	—	—	—	—	—
Magnet	—	1.04 [0.25, 4.24] (p=1.000)	2.53 [0.49, 13.17] (p=0.467)	—	—
Large Diameter (>2.5cm)	0.22 [0.04, 1.10] (p=0.070)	1.29 [0.43, 3.86] (p=0.857)	1.83 [0.62, 5.44] (p=0.413)	—	1.00 [0.32, 3.13] (p=1.000)

OR: Odds Ratio; CI: Confidence Interval; p: p-value. * indicates $p < 0.05$. Bold = statistically significant. — = missing or unstable estimate.

One death occurred in this subgroup [40], from 3,4-methylenedioxymethamphetamine (MDMA) induced acuphagia as the other motivation.

Although no statistically significant demographic differences were observed in this subgroup ($n = 9$ vs $n = 62$), there was a non-significant tendency toward more adults (+22%), individuals with prior ingestion history (+9%), those ingesting long objects (+9%), and those with severe disability (+8%). This group also showed lower prevalence of intent-to-harm motivation (–20%), fewer sharp object ingestions (–10%), and reduced psychiatric history and psychiatric motivation. This suggests a diverse subgroup.

Sharp object ingestion was associated with decreased odds of endoscopy. Subgroup analysis ($n = 34$ vs $n = 37$) revealed a significant reduction in large-diameter object ingestion (–11%) and magnet ingestion (–13%). There was a non-significant tendency to towards more conservative management (OR 1.51, 95% CI: 0.31-7.30, $p = 0.703$) and surgery (OR 2.27 95% CI: 0.85-6.06, $p = 0.157$).

All subgroup comparisons were conducted within the overall case-level dataset ($n = 71$). While observed differences may reflect genuine patterns in clinical behavior, they should be interpreted with caution due to the small sample sizes and potential for unmeasured confounding.

Subgroup analysis of predictors significantly associated with outcomes are shown graphically in Figure 5.

3.4.2 Meta-Analysis of Proportions

Pooled predictors and outcomes rates for the three included case series were first examined using meta-analysis of proportions, employing REML estimation with HK-adjusted 95% CIs. Results are presented in Figure 4.

All patients in the included case series were male and detained, resulting in a demographically homogenous cohort with respect to gender and detention status.

The pooled prevalence of psychiatric history was ≈40% (range: 0.1–99.8% across three studies), with substantial heterogeneity ($I^2 = 68\%$, $\tau^2 = 5.0$). Motivational subgroups showed moderate to high heterogeneity ($I^2 = 33$–69%). Protest motivation was reported in ≈78% of patients (range: 0–100%), with moderate heterogeneity ($I^2 = 32.7\%$, $\tau^2 = 4.3$). Intent-to-harm motivation was present in ≈4% of cases (range: 0–98%), with moderate heterogeneity ($I^2 = 40\%$, $\tau^2 = 4.0$). Psychiatric motivation was reported in ≈2% of cases (range: 0–100%), with high heterogeneity ($I^2 = 69\%$, τ^2: 3.6).

Meta-analysis was not feasible for several predictors—namely, displaced person status, alcohol use, severe disability, history of prior ingestion, large-diameter objects, and complications—as each was reported in only a single study. Similarly, meta-analysis of complication rates was not possible due to insufficient data.

Cases from case reports were collapsed and pooled for meta-analysis alongside the case series data. The results are shown in Fig 3.

Substantial heterogeneity was introduced in this process. Heterogeneity in detention status rose ($I^2 = 69\%$, $\tau^2 = 5$), as did gender heterogeneity.

Heterogeneity was only low in; psychiatric inpatient pooled proportions between 0–10% ($I^2 = 10$, τ^2 3); other motivation pooled proportion between 0–30% ($I^2 = 13$, $\tau^2 = 4.8$); and surgery pooled proportion of 10–60% ($I^2 = 17.5$, $\tau^2 = 0.8$).

These findings should be interpreted cautiously, as the wide confidence intervals and substantial heterogeneity likely reflect limited and inconsistent reporting across the small number of included studies.

3.4.3 Meta-Regression

Univariate meta-regression revealed exploratory associations between several predictors and the likelihood of surgical intervention across the three included studies. With this heterogeneity in mind, it was decided not to pool case series for univariate meta-regression, to examine detained case series as a subgroup against the more heterogenous case reports.

In detained populations, protest motivation was associated with a significantly lower likelihood of surgery (OR = 0.98, 95% CI: 0.98–0.98, $p = 0.003$). However, this result must be interpreted with caution due to the extremely small standard error and the limited number of studies. It is possible that this reflects quasi-complete separation (i.e. the outcome is nearly perfectly predicted by one variable) or instability in the regression estimates due to sparse data or model instability rather than a reliable effect.

Intent-to-harm motivation showed a positive association with surgery (OR = 1.29, 95% CI: 1.16–1.44, $p = 0.133$), though this was not statistically significant.

Male gender and prisoner status were both associated with a non-significant trend toward lower odds of surgery (OR = 0.97, 95% CI: 0.96–0.99, $p = 0.157$ for both).

Presence of a sharp object also trended toward reduced odds of surgery (OR = 0.95, 95% CI: 0.91–0.99, $p = 0.250$), but this was not statistically significant.

Neither psychiatric motivation (OR = 1.06, 95% CI: 0.96–1.18, $p = 0.448$) nor a history of psychiatric illness (OR = 1.03, 95% CI: 0.89–1.18, $p = 0.771$) showed a meaningful association with surgical intervention.

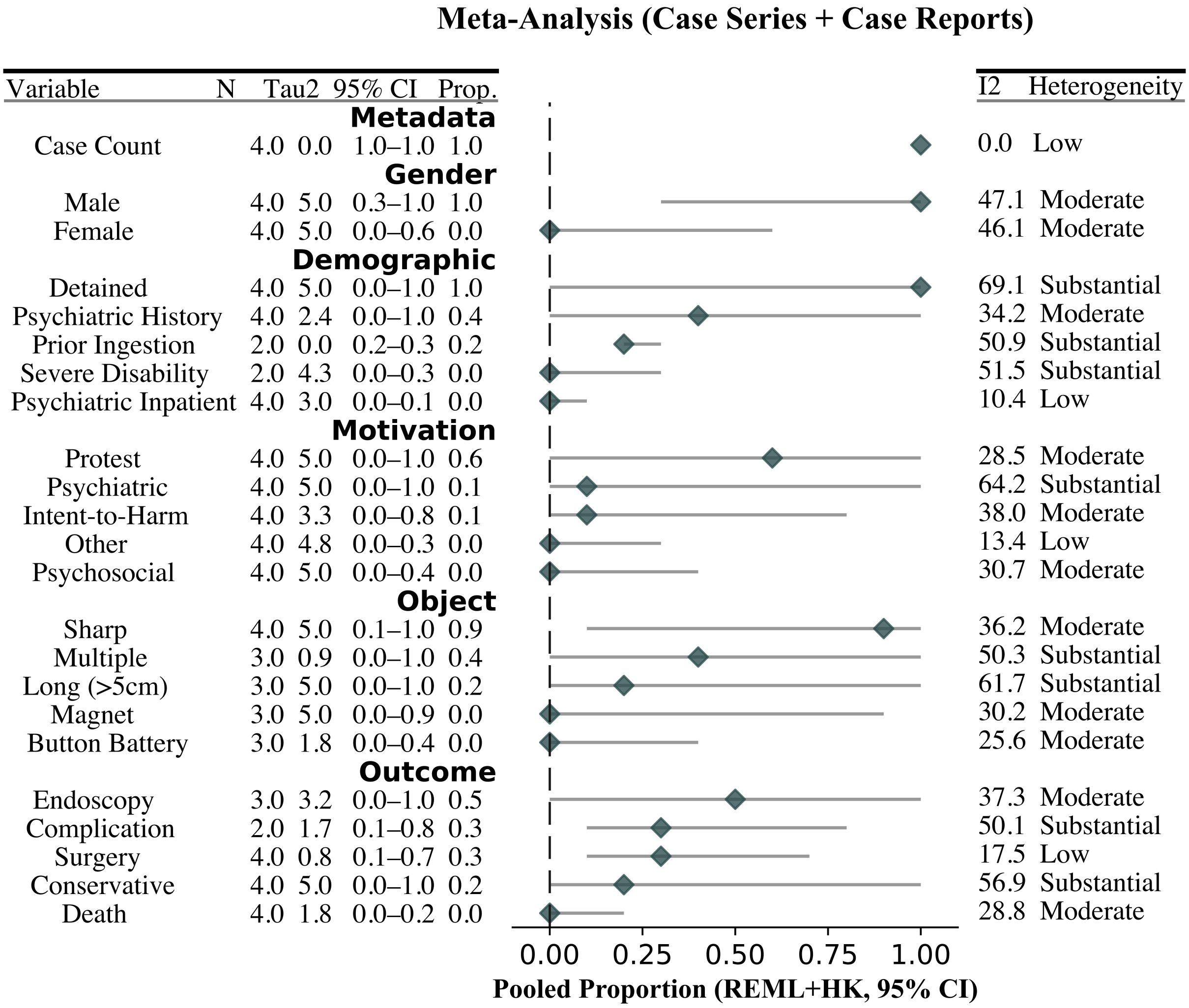

Fig. 3 Meta-Analysis of case series and pooled case reports. Meta-analysis of case series. Prop = Pooled Proportion; Tau2 = τ^2; N = n; I2 = I^2; REML and HK estimated 95% Confidence Intervals.

Given the limited number of series ($n = 3$), these findings should be regarded as exploratory, and interpreted with caution.

Results of univariate meta-regression are shown in Table 6.

3.5 Discussion

3.5.1 Intention is poorly reported

A total of 135 reports were excluded for not reporting intention and 91 were excluded for not reporting outcomes. The absence of these key variables meant that these studies could not be meaningfully integrated into the analysis. Collectively, these exclusions represent a substantial portion of the available literature and could have more than doubled the total sample size of this review. Their omission not only limits statistical power but also reflects a broader issue in the literature – namely, inconsistent or incomplete documentation of psychosocial factors and clinical outcomes. This under-reporting may introduce bias and restrict the generalisability of findings, particularly regarding the relationship between patient motivations and clinical decision-making.

3.5.2 Protest motivation, and intent-to-harm in detention, could modulate outcomes

Outcomes among detained individuals differed markedly across motivations. In case reports involving lesser-detained ingesters ($n = 71$), the pooled surgery rate was high (58%), with intent-to-harm as the dominant motivation (80%) and protest present in 55%. Sharp object ingestion was common (67%). Intent-to-harm was associated with nearly a six-fold increase in odds of surgery.

However, case series involving exclusively detained men ($n = 90$) in three countries revealed a contrasting trend. Despite pooled sharp object ingestion rates of approximately 90%, protest motivation was associated

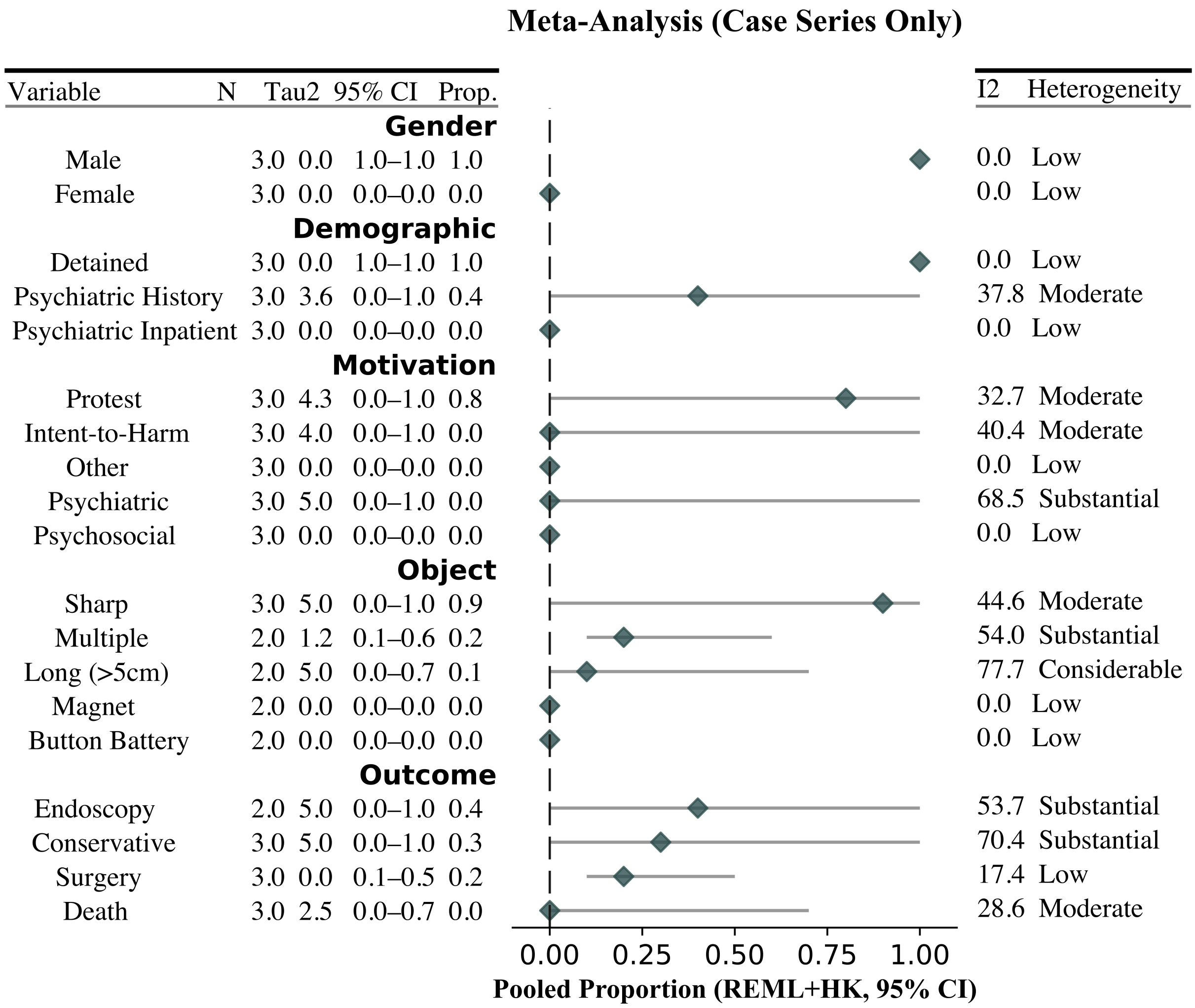

Fig. 4 Meta-Analysis of case series. Prop = Pooled Proportion; Tau2 = τ^2; N = n; I2 = I^2; REML and HK estimated 95% Confidence Intervals.

Table 6 Univariate meta-regression results (series-level).

Variable	Surgery
Demographic	
Detained	0.97 [0.96, 0.99] (p=0.157)
Psychiatric History	1.03 [0.89, 1.18] (p=0.771)
Gender	
Male	0.97 [0.96, 0.99] (p=0.157)
Motivation	
Intent-to-Harm	1.29 [1.16, 1.44] (p=0.133)
Protest	**0.98 [0.98, 0.98] (p=0.003)***
Psychiatric	1.06 [0.96, 1.18] (p=0.448)
Object	
Long (¿5cm)	—
Multiple	—
Sharp	0.95 [0.91, 0.99] (p=0.250)

OR: Odds Ratio; CI = 95% Confidence Interval.
* indicates $p < 0.05$; — = missing or unstable estimate.

with an 80% reduction in odds of surgery. This inverse association emerged despite moderate heterogeneity and wide confidence intervals.

In the largest series ($n = 52$) [73], protest motivation was nearly universal (97%), intent-to-harm was absent, and 64% of objects were sharp, yet the surgery rate was only 12%. In contrast, an older series ($n = 19$) [65] reported lower protest motivation (16%) and higher intent-to-harm (12%), with a surgery rate of 26%. Endoscopy rates were not reported in that series. A third series published in 2016 [39]—falling chronologically between the two—reported protest in 90%, sharp object ingestion in 84%, and an intent-to-harm rate of 10%, with a surgery rate of 21%, 79% managed conservatively, and one death.

These findings suggest that motivational context may shape clinical decisions, even when object risk is high. However, publication bias in case the case report cohort likely inflates surgery rates. For example, a Bulgarian series [77, 78] – reported in this review as cases due to the level of detail – described five inmates who ingested a total of 20 metal crosses – all required laparotomy for perforation.

3.5.3 Middle-age with differing motivations - reduced surgery but increased mortality

In case reports, individuals in the middle-aged group demonstrated a non-significant tendency towards reduced ingestion of sharp and large-diameter objects (−5% for both) and lower rates of intent-to-harm motivation. As intent-to-harm was associated with increased odds of surgery in this review, its lower prevalence may partly explain reduced surgical intervention in this group.

However, this apparent protective trend was offset by increased mortality. Two deaths occurred among middle-aged individuals, both outside custodial settings but with psychiatric motivations and history. One case involved a 54-year-old man with a 15-year history of repeated ingestions, who ultimately died after refusing surgery multiple times [69]. The other involved a patient who died after ingesting metal as a result of acuphagia linked to MDMA use [40].

This duality suggests middle age may represent a complex and vulnerable subgroup. On one side is a cohort with significant psychiatric comorbidities—reflected in higher recurrence—who may present late or decline treatment, thereby increasing mortality risk. On the other, a lower prevalence of intent-to-harm may contribute to reduced rates of surgical intervention.

3.6 Limitations

3.6.1 Publication bias, extraordinary cases and countertransference anger

Both of the above effects could be explained by publication bias, were deaths was published as they were exemplary and extraordinary cases.

This is demonstrated in the case reports. Some authors appear to favour sensational case reports, occasionally employing sarcastic or potentially inappropriate titles. Examples include: "Loose Screws", in reference to a female with an extensive psychiatric history who, "as advised by her rastafarian", ingested nails in a milkshake [51]; *Ingested Magnets: the force within* [114]; *Now You See It, Endo You Don't: Case of the Disappearing Knife* [42]; and *A Jackass And A Fish* [20]. Extreme cases of IIFO are remarkable, publication-worthy and interesting to read, likely leading to publication bias.

Furthermore, extreme cases of recurrent intentional ingestion cause "countertransference anger" [91] among clinicians who have to endure sometimes years of repeat encounters with prolific recurrent ingesters, who are often complex patients with complex mental health needs. This is well described by Gitlin et al. [46] who describes how clinicians are "held hostage" by ingesters "in the midst of their self-harming process" after an initiating an "ongoing injurious sequence". This anger, perhaps, finds an outlet in publication.

3.6.2 Selection Bias

This review was conducted by a single reviewer (JGE), which introduces the potential for selection bias, motivation miss-classification and inconsistent application of eligibility criteria. Although inclusion decisions were guided by a predefined protocol and supervision by an experienced reviewer (GC), dual independent screening was not performed due to resource limitations. This may have affected reproducibility and sensitivity of the search and screening processes.

3.6.3 Unmeasured confounding

The review did not record or analyse item location. This could have introduced an unmeasured confounding factor, perhaps reducing rates of perceived high risk ingestion, modulating outcome odds.

Time to presentation was not recorded. Ingesters that present early have increased successful endoscopy. This may have skewed endoscopy rates in some cases reports and case series, but the effect cannot be quantified in this review. It was attempted, but was difficult to standardise.

Also, this review did not record any alteration of objects that ingesters may have undertaken. Although, in clinical practice, this is likely difficult to assess. Sharp objects may wrapped to appear radiopaque and sharp on imaging, but in reality not be sharp or high risk, reducing the odds of morbidity [7, 14]. The opposite is also observed, as demonstrated in Bulgaria by the of home-made "gastrointestinal crosses". Paperclips were wrapped with paper and elastic and designed to spring open in the distal gastrointestinal tract. This lead to five perforations in five patients in one week in one prison [77]. Intention and motivation - in both examples - playing a clear role.

3.6.4 Information Bias

Information bias from missing data in studies likely distorted true effects in the case series analysis. This is particularly in age group and object characteristics analysis, but also endoscopic outcomes and demographic characteristics.

3.7 Conclusion

This review represents the first systematic synthesis of outcomes following intentional ingestion of foreign objects (IIFO) with a focus on patient motivation. Across both case reports and case series, the nature of the ingestion—particularly protest versus intent-to-harm—appears to modulate clinical decision-making, especially in detained populations. While protest-motivated ingestion was often managed conservatively, intent-to-harm cases were associated with significantly increased rates of surgical intervention.

Middle-aged adults (40–64 years) were significantly less likely to undergo surgery with a non-significant tendency to increased surgical and conservative management. Subgroup analysis suggests that psychiatric history, prior ingestion, and multiple object ingestion may contribute to more conservative management strategies and refusal of treatment, modulating this effect. Sharp object ingestion, traditionally viewed as high-risk, was not uniformly associated with surgical outcomes, particularly in the context of protest motivations and detention settings. However, outside of protest motivations in this age group, those with psychiatric co-morbidities had a tendency towards increased mortality.

Findings must be interpreted in light of limitations including small sample sizes, publication bias, selection bias, missing data, and unmeasured confounding factors such as object location, modification, and timing of presentation. Nonetheless, this review highlights the complexity of IIFO and the influence of both psychosocial context and clinical framing on treatment decisions.

Future research should focus on prospective, standardised reporting of IIFO cases to better understand how patient characteristics and motivations influence management. Greater clarity around risk stratification may help clinicians provide safer, more consistent care for this often-complex population.

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Statements and Declarations

The authors have no relevant financial or non-financial interests to disclose.

Data Availability

Data collection, manipulation and analysis in this review were conducted using *Python* [98] in *Visual Studio Code* [83] and *Jupyter Notebooks* [67]. The manuscript was compiled using *LaTeX* [71]. Specific *Python* packages used include: *Pandas* [117], *scikit-learn* [94], *statsmodel* [105], *seaborn* [129], *matplotlib* [57], *Cartopy* [81], *Forestplot* [107].

The data and code used in this systematic review are available on Github at https://github.com/jackgedge/iifo_systematic_review and on Queen Mary Research Online [44]. All appendices are supplied in the supplementary material.

Author Contributions

Conceptualisation: JGE, GC; Data curation: JGE; Formal analysis: JGE; Funding acquisition: JGE; Investigation: JGE; Methodology: JGE, GC; Project administration: JGE; Resources: JGE; Software: JGE; Supervision: GC; Validation: JGE, GC; Visualization: JGE; Writing – original draft: JGE; Writing – review & editing: JGE, GC.

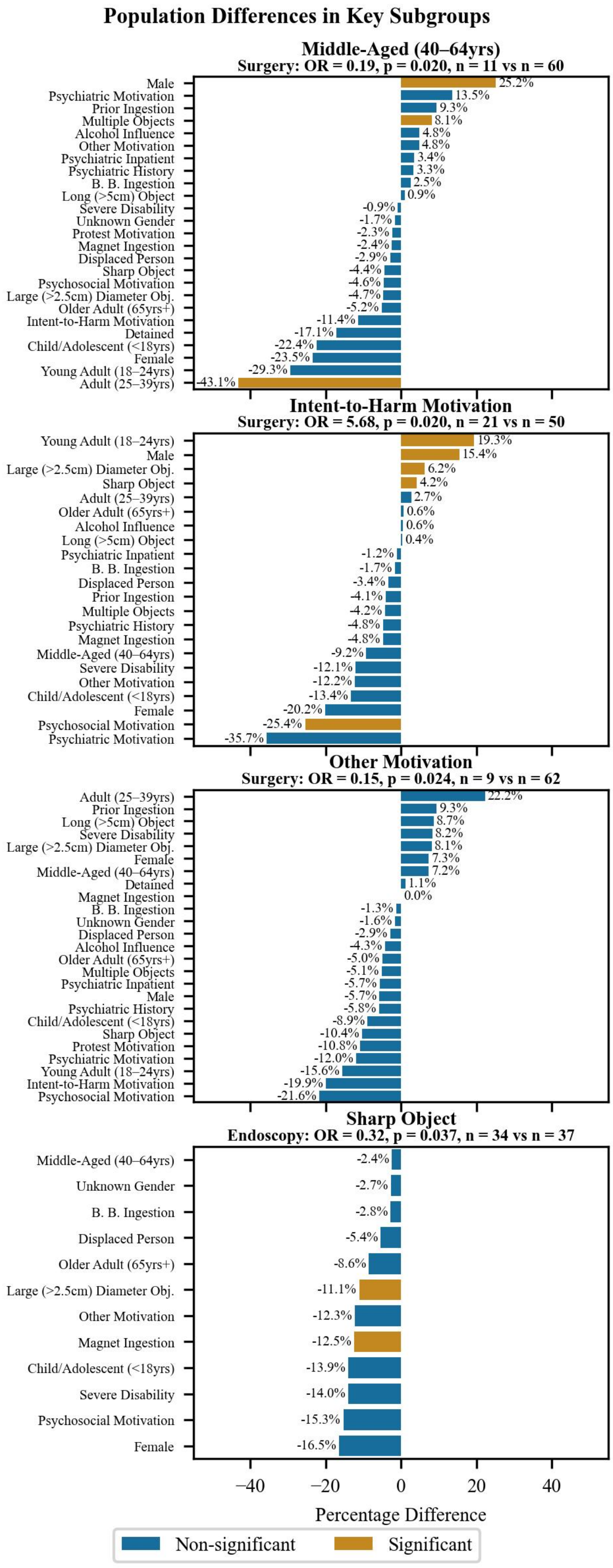

Fig. 5 Significant associated predictors from univariate analysis.

Supplementary Material

Appendix A Methods

A.1 Eligibility Criteria

Table A1 Inclusion criteria structured using the PICOS framework.

Category	**Details**
Population	Any human. Any age group.
Interventions or exposures	Humans that have: – Non-accidental ingestion – Ingestion of a true foreign body (non-nutritive items)
Comparators / Control group	Demographic: – Gender – Age – Detained Person – Psychiatric Inpatient - Displaced Person - Under Influence of Alcohol - Psychiatric History - Severely Disabled - Previous Ingestion Motivation: – Intent to harm – Psychiatric – Psychosocial – Protest – Other Object characteristics: – Button battery – Magnet – Long (>5 cm) – Large diameter (>2.5 cm) – Multiple – Blunt objects – Sharp-pointed objects
Outcomes of interest	– Endoscopic intervention – Surgical intervention – Conservative management – Complication rates – Mortality rates
Setting	Any setting.
Study designs	Any design.

Table A2 Exclusion criteria for study exclusion.

#	**Exclusion Criterion**
1	Full text not available in English.
2	Studies not focusing on intentional self-ingestion (into the gastrointestinal tract) of a foreign object via the oral cavity (mouth), or where it is unclear if ingestion occurred.
3	Studies focusing solely on accidental ingestion.
4	Non-human or animal studies.
5	Reviews, editorials, commentaries, and opinion pieces without original empirical data.
6	Duplicate publications or studies with overlapping datasets (only the most comprehensive or recent study was included).
7	Studies focusing on ingestion or co-ingestion of substances (e.g., poisons, medications) rather than physical foreign objects.
8	Ingestions undertaken in controlled environments as part of a voluntary study.
9	Ingestions not explicitly stated to be intentional and not suggestive of deliberate ingestion.
10	Does not meet inclusion criteria.
11	Ingestions where death resulted from other means (e.g., suicide by other method).
12	Studies published before the advent of endoscopy (1906).
13	Outcomes not reported.
14	Motivation not reported.
15	Object characteristics not reported

A.2 Search Strategy

Table A3 Concepts with associated keywords and MeSH terms used in PubMed search strategy.

Concept	Keywords	MeSH Terms
Foreign Bodies	“foreign obj*” “foreign bod*”	Foreign Bodies [MeSH]
Intentional Ingestion / Self-harm	“intent*” “deliberate*” “purpose*” “self-injur*” “selfharm*” “self-harm*”	Self-Injurious Behavior [MeSH]
Ingestion Behavior	“ingest*” “swallow*”	–
Interventions	“surg*” “endoscop*” “EGD” “OGD” “Esophagogastroduodenoscopy” “Oesophagogastroduodenoscopy” “manag*”	Endoscopy [MeSH] Surgical Procedures, Operative [MeSH] Conservative Treatment [MeSH] Drug Therapy [MeSH]

Table A4 Concepts with associated keywords and EMTREE terms used in Embase search strategy.

Concept	Keywords	EMTREE Terms
Foreign Bodies	“foreign obj*” “foreign bod*”	“foreign body”/exp
Intentional Ingestion / Self-harm	“intent*” “deliberate*” “purpose*” “self-injur*” “selfharm*” “self-harm*”	“automutilation”/exp
Ingestion Behavior	“ingest*” “swallow*”	“swallowing”/exp
Interventions	“surg*” “endoscop*” “EGD” “OGD” “Esophagogastroduodenoscopy” “Oesophagogastroduodenoscopy” “manag*”	“endoscopy”/exp “surgery”/exp ”conservative treatment’/exp ’drug therapy”/exp

Table A5 Concepts with associated keywords and Cochrane MeSH terms used in CENTRAL search strategy.

Concept	Keywords	Cochrane MeSH Terms
Foreign Bodies	"foreign obj*" "foreign bod*" (foreign NEXT obj*) (foreign NEXT bod*)	[mh foreign bodies]
Intentional Ingestion / Self-harm	intent* deliberate* purpose* (self NEXT injur*) (self NEXT harm*)	[mh self-injurious behavior]
Ingestion Behavior	ingest* swallow*	–
Interventions	surg* endoscop* EGD Esophagogastroduodenoscopy Oesophagogastroduodenoscopy manag*	[mh endoscopy] [mh surgical procedures, operative] [mh conservative treatment] [mh drug therapy]

Table A6 Concepts with associated keywords and Web of Science fields used in the search strategy.

Concept	Keywords	Search Field
Foreign Bodies	foreign obj* foreign bod*	ALL=
Intentional Ingestion / Self-harm	automutilation intent* deliberate* purpose* self-injur* selfharm* self-harm*	ALL=
Ingestion Behavior	swallowing ingest* swallow*	ALL=
Interventions	endoscopy surgery conservative treatment drug therapy surg* endoscop* EGD Esophagogastroduodenoscopy Oesophagogastroduodenoscopy manag*	ALL=

Table A7 Concepts with associated keywords and Scopus syntax used in the search strategy.

Concept	Keywords	Search Field / Syntax
Foreign Bodies	foreign PRE/0 obj* foreign PRE/0 bod*	ALL()
Intentional Ingestion / Self-harm	intent* deliberate* purpose* self PRE/0 injur* self PRE/0 harm*	ALL()
Ingestion Behavior	ingest* swallow*	ALL()
Interventions	endoscopy surgery 'conservative' 'treatment' 'drug' 'therapy' surg* endoscop* egd esophagogastroduodenoscopy oesophagogastroduodenoscopy manag*	ALL()

Table A8 Concepts with associated keywords and controlled vocabulary (Descriptors) used in PsycINFO search strategy.

Concept	Keywords	PsycINFO Descriptors
Foreign Bodies	foreign obj* foreign bod*	–
Intentional Ingestion / Self-harm	automutilation intent* deliberate* purpose* self injur* self harm*	DE "Nonsuicidal Self-Injury"
Ingestion Behavior	ingest* swallow*	DE "Ingestion"
Interventions	endoscop* conservative treatment drug therapy surg* egd esophagogastroduodenoscopy oesophagogastroduodenoscopy manag*	DE "Surgery"

Table A9 Concepts with associated keywords used in Google Scholar search strategy.

Concept	Keywords	Search Field
Foreign Bodies	"foreign obj*" "foreign bod*"	–
Intentional Ingestion / Self-harm	"intent*" "deliberate*" "purpose*" "self-injur*" "selfharm*" "self-harm*"	–
Ingestion Behavior	"ingest*" "swallow*"	–

A.2.1 PubMed Query

("Foreign Bodies"[MeSH] OR "foreign obj*" OR "foreign bod*")
AND
("Self-Injurious Behavior"[MeSH] OR "intent*" OR "deliberate*" OR "purpose*" OR "self-injur*" OR "self-harm*" OR "self-harm*")
AND
("ingest*" OR "swallow*")
AND
("Endoscopy"[MeSH] OR "Surgical Procedures, Operative"[MeSH] OR "Conservative Treatment"[MeSH] OR "Drug Therapy"[MeSH] OR "surg*" OR "endoscop*" OR "EGD" OR "OGD" OR "Esophagogastroduodenoscopy" OR "Oesophagogastroduodenoscopy" OR "manag*")

A.2.2 Embase Query (All Fields)

('foreign body'/exp OR "foreign obj*" OR "foreign bod*")
AND
('automutilation'/exp OR "intent*" OR "deliberate*" OR "purpose*" OR "self-injur*" OR "selfharm*" OR "self-harm*")
AND
('swallowing'/exp OR "ingest*" OR "swallow*")
AND
('endoscopy'/exp OR 'surgery'/exp OR 'conservative treatment'/exp OR 'drug therapy'/exp OR "surg*" OR "endoscop*" OR "EGD" OR "OGD" OR "Esophagogastroduodenoscopy" OR "Oesophagogastroduodenoscopy" OR "manag*")

A.2.3 CENTRAL (Cochrane) Query (All Fields)

([mh "foreign bodies"] OR ("foreign" NEXT "obj*") OR ("foreign" NEXT "bod*"))
AND
([mh "self-injurious behavior"] OR "intent*" OR "deliberate*" OR "purpose*" OR ("self" NEXT "injur*") OR ("self" NEXT "harm*"))
AND
("ingest*" OR "swallow*")
AND
([mh "endoscopy"] OR [mh "surgical procedures, operative"] OR [mh "conservative treatment"] OR [mh "drug therapy"] OR "surg*" OR "endoscop*" OR "EGD" OR "Esophagogastroduodenoscopy" OR "Oesophagogastroduodenoscopy" OR "manag*")

A.2.4 Web of Science Query

Link: https://www.webofscience.com/wos/woscc/summary/4da44d48-3e09-4a94-a3bd-ff8139e94859-01387ccd63/relevance/1

ALL=("foreign obj*" OR "foreign bod*")
AND
ALL=("automutilation" OR "intent*" OR "deliberate*" OR "purpose*" OR "self-injur*" OR "selfharm*" OR "self-harm*")
AND
ALL=("swallowing" OR "ingest*" OR "swallow*")
AND
ALL=("endoscopy" OR "surgery" OR "conservative treatment" OR "drug therapy" OR "surg*" OR "endoscop*" OR "EGD" OR "Esophagogastroduodenoscopy" OR "Oesophagogastroduodenoscopy" OR "manag*")

A.2.5 Scopus Query

ALL ("foreign PRE/0 obj*" OR "foreign PRE/0 bod*")
AND
ALL ("intent*" OR "deliberate*" OR "purpose*" OR "self PRE/0 injur*" OR "self PRE/0 harm*")
AND
ALL ("ingest*" OR "swallow*")
AND
ALL ("endoscopy" OR "surgery" OR "conservative" OR "treatment" OR "drug" OR "therapy" OR "surg*" OR "endoscop*" OR "EGD" OR "Esophagogastroduodenoscopy" OR "Oesophagogastroduodenoscopy" OR "manag*")

A.2.6 PsycINFO Query

Link: https://search-ebscohost-com.ezproxy.library.qmul.ac.uk/login.aspx?direct=true&db=psyh&bquery=(foreign+obj*+OR+foreign+bod*)+AND+(DE+%26quot%3bNonsuicidal+Self-Injury%26quot%3b+OR+automutilation+OR+intent*+OR+deliberate*+OR+purpose*+OR+self+injur*+OR+self+harm*)+AND+(DE+%26quot%3bIngestion%26quot%3b+OR+ingest*+OR+swallow*)+AND+(DE+%26quot%3bSurgery%26quot%3b+OR+endoscop*+OR+conservative+treatment+OR+drug+therapy+surg*+OR+endoscop*+OR+egd+OR+esophagogastroduodenoscopy+OR+oesophagogastroduodenoscopy+OR+manag*)&type=0&searchMode=Standard&site=ehost-live

(“foreign obj*” OR “foreign bod*”)
AND
(DE “Nonsuicidal Self-Injury” OR “automutilation” OR “intent*” OR “deliberate*” OR “purpose*” OR “self injur*” OR “self harm*”)
AND
(DE “Ingestion” OR “ingest*” OR “swallow*”)
AND
(DE “Surgery” OR “endoscop*” OR “conservative treatment” OR “drug therapy” OR “surg*” OR “endoscop*” OR “EGD” OR “Esophagogastroduodenoscopy” OR “Oesophagogastroduodenoscopy” OR “manag*”)

A.3 Grey Literature

Google Scholar

(“foreign obj*” OR “foreign bod*”)
AND
(“intent*” OR “deliberate*” OR “purpose*” OR “self-injur*” OR “selfharm*” OR “self-harm*”)
AND
(“ingest*” OR “swallow*”)

A.4 Screening Process

A total of 808 records were identified through initial database searches: PubMed (317), Web of Science (277), Google Scholar (135), Embase (25), SCOPUS (24), PsycINFO (16), and Cochrane (14). 316 duplicates were identified and removed.

Title and abstract screening was undertaken, with JGE reviewing all 492 records. A random sample of 50 records was generated for independent screening MS. Cohen's Kappa was calculated for inter-reviewer agreement between JGE and MS, yielding a value of 0.38, indicating fair agreement. Where JGE and MS disagreed, 16 records were reviewed by GC. In total, 176 records were excluded, leaving 316 for full text review.

During full text review, JGE reviewed all 316 records. A random sample of 32 records was generated for independent review by MS. Inter-reviewer agreement was again calculated using Cohen's Kappa, yielding a value of 0.21, indicating fair agreement. Where JGE and MS disagreed, 5 records were reviewed by GC. In total, 276 records were excluded during full text review. 40 records were included and proceeded to bibliography search.

The bibliographies of the 40 included papers were searched by manually JGE. Relevant bibliography items were identified, collated, and evaluated against the eligibility criteria, yielding 194 results.

These 194 results were reviewed by JGE. 164 bibliography search records were excluded, leaving 30 for inclusion.

Therefore, a total of 70 records were included in this study and proceeded to bias assessment. This process is illustrated in Figure 1.

A.5 Data Extraction

A.5.1 Definitions

Full variable definitions are shown in A10.

For the purposes of this study, "surgery" was defined as "any operative intervention performed in a sterile operating theatre under general or regional anaesthesia, involving incision or surgical access to body cavities (including laparotomy, laparoscopy, thoracotomy, or cervical exploration) for the purpose of removing an ingested object or managing complications of ingestion". Procedures performed "solely via flexible or rigid endoscopy through natural orifices" were categorised as "endoscopy" and not considered surgical interventions.

A.5.2 Process

Data were initially extracted by a single reviewer (JGE) into *Microsoft Excel* [82]. Variables for extraction were developed iteratively through engagement with the literature and analysis of consistent reporting patterns. A preliminary review of the first 30 case reports informed the development of additional data categories, which were subsequently applied to the remaining reports.

Following initial extraction, data were imported into *Python* [98] for further processing and analysis. The Python-based pipeline included data cleaning, validation, and transformation to ensure consistency across heterogeneous study formats. These structured data were then used to guide the extraction of aggregate data from case series. Studies were grouped for extraction based on their classification as case reports or case series. Where case series contained sufficiently granular data, cases were extracted individually and treated as case reports; otherwise, data were extracted at the aggregate level. Case grouping for analysis followed the criteria for inclusion as individual case reports or case series, as defined above. Relevant data from reviews and other literature types were recorded under the case report category.

A.6 Computational Risk of Bias Assessment

To reduce bias dilution of intentionality effect, a novel computation risk of bias assessment was undertaken, using a combination of human review followed by computional risk of bias assessment. First, the author (JGE) extracted data into Microsoft Excel [82]. Then, a computation risk of bias filter was applied to extracted case report and case series data. That process is outlined in this appendix.

A.6.1 Case Reports

For case reports, the JBI Checklist for Case Reports was used. This tool assesses eight domains of reporting quality, including whether patient demographics were clearly described, a timeline of clinical history was provided, the presenting condition and diagnostic assessment were outlined, and whether the intervention, post-intervention condition, and any adverse events were reported. The final domain evaluates whether the case provides meaningful takeaway lessons.

Table A10 Variables used for case report data extraction. Aggregates of which where used to create Variable_Rate and Variable_Cases.

Variable	Definition
Is_Prisoner	Documented in prison, police custody, or detained (including immigration detention) at the time of the encounter; 'N' if not detained; 'UK' if unknown.
Psych_Hx	Documented DSM-V mental disorder (including substance-related disorders) [11]; 'N' if no diagnosis; 'UK' if data unavailable.
Is_Displaced_Person	'Y' if: meets the UN General Assembly [122] definition of 'Refugee'; or meets UNHCR [37] definition of an 'internally displaced person'; or meets the UNHCR [125] definition for 'asylum seeker'; 'N' if not displaced; 'UK' if unknown.
Under_Influence_Alcohol	Evidence, suspicion, or self-report of alcohol influence at presentation; 'N' if no indication; 'UK' if unknown.
Is_Psych_Inpat	Admitted (voluntarily or involuntarily) to a psychiatric facility/ward at encounter; 'N' if not admitted; 'UK' if unknown.
Severe_Disability_Hx	History of severe learning disability or impaired consciousness; 'N' if absent; 'UK' if unknown.
Previous_Ingestions	Prior episode of foreign-body ingestion documented; 'N' if first ingestion; 'UK' if history unknown.
Motivation_Intent_To_Harm	Ingestion intended for self-harm, self-injury, or suicide; 'N' if other motive; 'UK' if unclear.
Motivation_Protest	Ingestion as protest, demonstration, or manipulation (e.g., objection to detention conditions); 'N' if not protest-related; 'UK' if unclear.
Motivation_Psychiatric	Ingestion driven primarily by an underlying psychiatric condition (psychosis, impulsivity, etc.); 'N' if not psychiatric; 'UK' if unclear.
Motivation_Psychosocial	Ingestion motivated by social or interpersonal factors (imitative acts, shock value, body-image, safekeeping, etc.); 'N' if not psychosocial; 'UK' if unclear.
Motivation_Unknown	No clear motivation identified in documentation; 'N' if specific motive recorded; 'UK' if ambiguous.
Object_Button_Battery	Button battery ingested; 'N' if not; 'UK' if object type not recorded.
Object_Magnet	Magnet ingested; 'N' if none; 'UK' if unknown.
Object_Long	Ingested object length > 5 cm; 'N' if ≤ 5 cm; 'UK' if dimensions unknown.
Object_Sharp	Object described as sharp or pointed (e.g., blades, nails, needles); 'N' if not sharp; 'UK' if unclear.
Object_Multiple	More than one object ingested in same episode; 'N' for single object; 'UK' if number unspecified.
Object_Unknown	Where object characteristics are unknown. 'N' if known; 'UK' if Unknown.
Outcome_Endoscopy	Endoscopic intervention performed during episode; 'N' if not; 'UK' if unavailable.
Outcome_Surgery	Surgical intervention performed (operative procedure under anaesthesia); 'N' if not; 'UK' if not documented.
Outcome_Conservative	'Y' if managed without endoscopy or surgery; 'N' if either procedure performed.
Outcome_Death	Death causally related to ingestion complications; 'N' if survived; 'UK' if outcome unknown.
Outcome_Complication	'Y' if any complication directly related to ingestion or resulting from management strategy; 'N' if no complication; 'UK' if unknown.
Outcome_Unknown	Where no outcome identified; 'N' if outcome identified; 'UK' if Unknown.

In addition to manual JBI appraisal, a logic-based validation filter was applied to all case reports using *Python Pandas* [117]. This secondary filter assessed whether key variables — specifically, outcomes, object characteristics, and motivation — were completely unreported. For each domain, a binary flag was generated:

- *Outcome_Unknown* was marked `1` if all outcome-related fields were either missing or marked as unknown.
- *Object_Unknown* was marked `1` if all object-related fields (excluding *Object_Other_Long*) were missing or unknown.
- *Motivation_Unknown* was predefined in the dataset and indicated absence of motivational information.

If any of these flags were triggered, the corresponding JBI item most affected by the missing domain was marked as not reported (e.g., *Post_Intervention_Condition_Described* or *History_Timeline* set to `N`). Finally, an *Overall_Appraisal* score of *Exclude* was assigned, indicating high risk of bias and exclusion from analysis. This ensured that only case reports with sufficient information to meaningfully contribute to the review question were retained.

A.6.2 Case Series

For case series, the JBI Checklist for Case Series was applied. The JBI Checklist for Case Series assesses 10 domains of methodological and reporting quality. These include whether the case series defined clear inclusion criteria, applied valid and consistent methods to identify the condition, and included participants consecutively and completely. The checklist also evaluates whether participant demographics and clinical information were clearly reported, whether outcomes or follow-up results were adequately described, and whether the study setting was detailed. Finally, it considers whether the statistical analysis used was appropriate for the data presented.

In addition to manual JBI appraisal, a logic-based exclusion filter was applied using *Python Pandas* [117]. This filter assessed whether key variables — specifically, motivation, object characteristics, and outcomes — were unreported for the entire study population. For each of these domains, a derived rate variable was calculated:

- *Outcome_Unknown_Rate* was marked as `1` if all outcome-related fields were missing or marked as unknown (i.e. the entire population had an had an unknown outcome).
- *Motivation_Unknown_Rate* indicated whether motivation was absent or only partially reported across cases within the study.
- *Object_Unknown_Rate* was derived if all object-related fields were missing or unknown.

If any of these indicators were flagged, the corresponding JBI checklist item (e.g., *Clear_Outcome_Followup_Reported*, *Clear_Demographic_Reporting*, or *Clear_Clinical_Info_Reporting*) was marked as `N`, and the study received an *Overall_Appraisal* of `Exclude`. This logic-based validation ensured that case series lacking essential variables could be systematically excluded from the final analysis, maintaining consistency with the review question and minimising risk of bias in the dataset.

Appendix B PRISMA Checklist

PRISMA

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title Page
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Introduction
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Introduction
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Methods
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Methods
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Methods & Appendix
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Methods & Appendix
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Methods & Appendix
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Methods & Appendix
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Methods & Appendix
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Methods
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Methods & Appendix
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Methods & Appendix & Github
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Methods
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Methods
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Methods
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Methods

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Methods
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Methods & Results
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	PRISMA Diagram
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Results
Study characteristics	17	Cite each included study and present its characteristics.	Results
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Results
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Results
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Results
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Results
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Results
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Results
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Results
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Results
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Discussion
	23b	Discuss any limitations of the evidence included in the review.	Dsicussion
	23c	Discuss any limitations of the review processes used.	Discussion
	23d	Discuss implications of the results for practice, policy, and future research.	Discussion
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Methods
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Methods
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Github
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Funding
Competing interests	26	Declare any competing interests of review authors.	Disclosure

PRISMA

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Data Availability

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.

References

[1] Aitchison G, Essex R (2024) Self-harm in immigration detention: political, not (just) medical. J Med Ethics 50(11):786–793. https://doi.org/10.1136/jme-2022-108366

[2] Ajdacic-Gross V, Weiss MG, Ring M, et al (2008) Methods of suicide: international suicide patterns derived from the who mortality database. Bulletin of the World Health Organization 86(9):726. https://doi.org/10.2471/BLT.07.043489

[3] Akay S, Günay S, Binicier ÖB, et al (2015) A deliberately swallowed foreign body: money package. Endoscopy 47 Suppl 1:E602–603. https://doi.org/10.1055/s-0035-1569668

[4] Al-Faham FSM, Al-Hakkak SMM (2020) The largest esophageal foreign body in adults: A case report. Ann Med Surg 54:82–84. https://doi.org/10.1016/j.amsu.2020.04.039

[5] Al Shaaibi R, Al Waili I (2021) Laparoscopic retrieval of ingested foreign body. Oman Med J 36(3):e264. https://doi.org/10.5001/omj.2021.35

[6] Alao AO, Abraham B (2006) Foreign body ingestions in a schizophrenic patient. West Afr J Med 25(3):239–241. https://doi.org/10.4314/wajm.v25i3.28286

[7] Albeldawi M, Birgisson S (2014) Conservative management of razor blade ingestion. Gastroenterol Rep 2(2):158–159. https://doi.org/10.1093/gastro/gou002

[8] Ali A, Alhindi S (2020) A child with a gastrocolic fistula after ingesting magnets: An unusual complication. Cureus 12(7):e9336. https://doi.org/10.7759/cureus.9336

[9] Ali A, Mahgoub AM, Emad S, et al (2022) Endoscopic retrieval of an ingested mobile phone from the stomach of a prisoner: When gastroenterologists answer the call. Cureus 14(12):e33053. https://doi.org/10.7759/cureus.33053

[10] Ali SM (2017) Duodenal perforation by swallowed toothbrush: Case report and review of literature. Open Access J Surg 4(2). https://doi.org/10.19080/OAJS.2017.04.555632

[11] American Psychiatric Association (2013) Diagnostic and statistical manual of mental disorders (5th ed.). URL https://psychiatryonline.org/doi/book/10.1176/appi.books.9780890425787

[12] Amnesty International (2024) Refugees, asylum seekers and migrants. URL https://www.amnesty.org/en/what-we-do/refugees-asylum-seekers-and-migrants/

[13] Apikotoa S, Ballal H, Wijesuriya R (2022) Endoscopic foreign body retrieval from the caecum - a case report and push for intervention guidelines. Int J Surg Case Rep 90:106755. https://doi.org/10.1016/j.ijscr.2022.106755

[14] Ataya A, Alraiyes A, Alraies M (2013) Razor blades in the stomach. QJM: An Int J Med 106(8):783–784. https://doi.org/10.1093/qjmed/hcs165

[15] Atayan Y, Cagin YF, Erdogan MA, et al (2016) Lighter ingestion as an uncommon cause of severe vomiting in a schizophrenia patient. Case Rep Gastrointest Med 2016:6301302. https://doi.org/10.1155/2016/6301302

[16] Athwal H (2015) 'i don't have a life to live': deaths and uk detention. Race & Class 56(3):50–68. https://doi.org/10.1177/0306396814556224

[17] Barros JL, Caballero A, Rueda JC, et al (1991) Foreign body ingestion: management of 167 cases. World J Surg 15(6):783–788. https://doi.org/10.1007/BF01665320

[18] Becq A, Camus M, Dray X (2021) Foreign body ingestion: dos and don'ts. Frontline Gastroenterol 12(7):664–670. https://doi.org/10.1136/flgastro-2020-101450

[19] Beecroft N, Bach L, Tunstall N, et al (1998) An unusual case of pica. Int J Geriatr Psychiatry 13(9):638–641. https://doi.org/10.1002/(sici)1099-1166(199809)13:9⟨638::aid-gps837⟩3.0.co;2-n

[20] Benoist LBL, van der Hoven B, de Vries AC, et al (2019) A jackass and a fish: A case of life-threatening intentional ingestion of a live pet catfish (corydoras aeneus). Acta Oto-Laryngol Case Rep https://doi.org/10.1080/23772484.2018.1555436

[21] Berry P, Kotha S (2022) Crying wolf: the danger of recurrent intentional foreign body ingestion. Frontline Gastroenterol 13(3):266. https://doi.org/10.1136/flgastro-2021-101888

[22] Bevione F, Panero M, Abbate-Daga G, et al (2024) Risk of suicide and suicidal behavior in refugees. a meta-review of current systematic reviews and meta-analyses. J Psychiatr Res 177:287–298. https://doi.org/10.1016/j.jpsychires.2024.07.024

[23] Bhasin SK, Kachroo SL, Kumar V, et al (2014) 7" long knife for 7 years in the duodenum: a rare case report and review of literature. Int Surg J 1(1):29–32

[24] Bhattacharjee P, Singh O (2008) Repeated ingestion of sharp-pointed metallic objects. Arch Iran Med 11:563–5

[25] Bhugra D, Craig TKJ, Bhui K (2010) Mental Health of Refugees and Asylum Seekers. OUP Oxford

[26] Bhumi S, Mago S, Mavilia-Scranton MG, et al (2024) Esophageal button battery retrieval: Time-in may not be everything. Cureus 16(4):e58327. https://doi.org/10.7759/cureus.58327

[27] Birk M, Bauerfeind P, Deprez P, et al (2016) Removal of foreign bodies in the upper gastrointestinal tract in adults: European society of gastrointestinal endoscopy (esge) clinical guideline. Endoscopy 48(05):489–496. https://doi.org/10.1055/s-0042-100456

[28] Bozorgmehr R, Bahadorinia M, Pouyanfar S, et al (2022) A rare case of abdominal foreign bodies; laparoscopic removal of a sewing needle. Ann Med Surg 82:104747. https://doi.org/10.1016/j.amsu.2022.104747

[29] Buils F, Renau G, Sánchez Cano J, et al (2024) Repeated behavior of deliberate foreign body ingestion in a patient with psychiatric disorder. a case report. J Med Res Surg https://doi.org/10.52916/jmrs244144

[30] Camacho Dorado C, Sánchez Gallego A, Miota de Llama JI, et al (2018) Metallic bezoar after suicide attempt. Cir Esp 96(8):515. https://doi.org/10.1016/j.ciresp.2018.02.015

[31] Cauchi JA, Shawis RN (2002) Multiple magnet ingestion and gastrointestinal morbidity. Arch Dis Child 87(6):539–540. https://doi.org/10.1136/adc.87.6.539

[32] Chalk SG (1928) Foreign bodies in the stomach: Report of a case in which more than two thousand five hundred foreign bodies were found. Arch Surg 16(2):494. https://doi.org/10.1001/archsurg.1928.01140020045003

[33] Chang WJ, Chiu WY (2017) Gastric foreign body: a comb. Clin Case Rep 5(6):1036–1037. https://doi.org/10.1002/ccr3.957

[34] Cox D, Donohue P, Costa V (2007) A swallowed toothbrush causing perforation 2 years after ingestion. Br J Hosp Med 68(10):559. https://doi.org/10.12968/hmed.2007.68.10.27330

[35] Csáky G, Szederkényi I, Botos A, et al (1998) Laparoscopic removal of a foreign body from the jejunum. Surg Laparosc Endosc 8(1):68–70

[36] Davidson H, Doherty B (2016) Iranian refugee critically ill after setting himself on fire on nauru during un visit. URL https://www.theguardian.com/australia-news/2016/apr/27/iranian-asylum-seeker-severely-burnt-after-setting-himself-on-fire-on-nauru

[37] Deng FM (1998) Guiding Principles on Internal Displacement. UN Doc E/CN.4/1998/53/Add.2, United Nations, URL https://digitallibrary.un.org/record/535577?v=pdf

[38] Divsalar P, Hosseini Mousa S, Hayatbakhsh Abbasi M (2023) Repeated intentional swallowing of foreign objects by an adolescent girl (case report). Int J High Risk Behav Addict https://doi.org/10.5812/ijhrba-134720

[39] Elghali MA, Ghrissi R, Fadhl H, et al (2016) The management of voluntary ingestion of razor blades by inmates. Int Surg 105(1-3):129–133. https://doi.org/10.9738/INTSURG-D-16-00204.1

[40] Emamhadi MA, Najari F, Hedayatshode MJ, et al (2018) Sudden death following oral intake of metal objects (acuphagia): a case report. Emerg 6(1):e16. URL https://www.researchgate.net/publication/326447088_Sudden_Death_Following_Oral_Intake_of_Metal_Objects_Acuphagia_a_Case_Report

[41] Farhadi F, Mohtadi A, Pakmehr M, et al (2024) This is a successful removal of more than 450 pieces of metal objects from a patient's stomach: a case report. J Med Case Rep 18(1):381. https://doi.org/10.1186/s13256-024-04672-3

[42] Fine S, Watson JB, Habr F (2013) Now you see it, endo you don't: Case of the disappearing knife. Gastroenterol 144(7):e6–e7. https://doi.org/10.1053/j.gastro.2013.01.059

[43] Fry E, Counselman FL (2010) A right scrotal abscess and foreign body ingestion in a schizophrenic patient. J Emerg Med 38(5):587–592. https://doi.org/10.1016/j.jemermed.2007.07.018

[44] Galbraith-Edge J, Cattermole G (2025) Does Motivation Matter? A systematic review and meta-analysis of outcomes following intentional ingestion of foreign objects. https://doi.org/10.17636/101113411, URL https://qmro.qmul.ac.uk/xmlui/handle/123456789/113411

[45] Gardner A, Radwan R, Allison M, et al (2017) Double duodenal perforation following foreign body ingestion. BMJ Case Rep 2017. https://doi.org/10.1136/bcr-2017-223182

[46] Gitlin DF, Caplan JP, Rogers MP, et al (2007) Foreign-body ingestion in patients with personality disorders. Psychosomatics 48(2):162–166. https://doi.org/10.1176/appi.psy.48.2.162

[47] Global Detention Project (2024) United kingdom immigration detention profile. URL https://www.globaldetentionproject.org/countries/europe/united-kingdom

[48] Goldman RD, Schachter P, Katz M, et al (1998) A bizarre bezoar: case report and review of the literature. Paediatr Surg Int 14(3):218–219. https://doi.org/10.1007/s003830050492

[49] Guinan D, Drvar T, Brubaker D, et al (2019) Intentional foreign body ingestion: A complex case of pica. Case Rep Gastroenterol Med 2019:7026815. https://doi.org/10.1155/2019/7026815

[50] Haase E, Schönfelder A, Nesterko Y, et al (2022) Prevalence of suicidal ideation and suicide attempts among refugees: a meta-analysis. BMC Public Health 22(1):635. https://doi.org/10.1186/s12889-022-13029-8

[51] Hardy JC, Ashcroft C, Kay C, et al (2023) Loose screws: Removal of foreign bodies from the lower gastrointestinal tract. Cureus 15(8):e43093. https://doi.org/10.7759/cureus.43093

[52] Hedrick K, Borschmann R (2023) Self-harm among unaccompanied asylum seekers and refugee minors: protocol for a global systematic review of prevalence, methods and characteristics. BMJ Open 13(6):e069237. https://doi.org/10.1136/bmjopen-2022-069237

[53] Hedrick K, Armstrong G, Coffey G, et al (2019) Self-harm in the australian asylum seeker population: A national records-based study. SSM Popul Health 8:100452. https://doi.org/10.1016/j.ssmph.2019.100452

[54] Higgins J, Thomas J, Chandler J, et al (2024) Cochrane Handbook for Systematic Reviews of Interventions version 6.5 (updated August 2024). Cochrane, URL https://www.cochrane.org/handbook

[55] Hsieh A, Hsiehchen D, Layne S, et al (2020) Trends and clinical features of intentional and accidental adult foreign body ingestions in the united states, 2000 to 2017. Gastrointest Endosc 91(2):350–357.e1. https://doi.org/10.1016/j.gie.2019.09.010

[56] Hunt I, Hartley S, Alwahab Y, et al (2007) Aortoesophageal perforation following ingestion of razor-blades with massive haemothorax. Eur J Cardiothorac Surg 31(5):946–948. https://doi.org/10.1016/j.ejcts.2007.01.073

[57] Hunter JD (2007) Matplotlib: A 2d graphics environment. Comput Sci Eng 9(3):90–95. https://doi.org/10.1109/MCSE.2007.55

[58] Ikenberry SO, Jue TL, Anderson MA, et al (2011) Management of ingested foreign bodies and food impactions. Gastrointest Endosc 73(6):1085–1091. https://doi.org/10.1016/j.gie.2010.11.010

[59] Jackson CL (1957) Foreign bodies in the esophagus. Am J Surg 93(2):308–312. https://doi.org/10.1016/0002-9610(57)90783-3

[60] Jaini PA, Haliburton J, Rush AJ (2023) Management challenges of recurrent foreign body ingestions in a psychiatric patient: A case report. J Psychiatr Pract 29(2):167–173. https://doi.org/10.1097/PRA.0000000000000694

[61] Jehangir M, Mallory C, Medverd JR (2019) Digital tomosynthesis for detection of ingested foreign objects in the emergency department: a case of razor blade ingestion. Emerg Radiol 26(2):249–252. https://doi.org/10.1007/s10140-018-01664-x

[62] Jin S, Horiguchi T, Ma X, et al (2023) Metallic foreign bodies ingestion by schizophrenic patient: a case report. An Med Surg 85(4):1270. https://doi.org/10.1097/MS9.0000000000000497

[63] Kar SK, Kamboj A, Kumar R (2015) Pica and psychosis - clinical attributes and correlations: a case report. J Fam Med Pri Care 4(1):149–150. https://doi.org/10.4103/2249-4863.152277

[64] Kariholu PL, Jakareddy R, Hemanthkumar M, et al (2008) Pica - a case of acuphagia or hyalophagia? Indian J Surg 70(3):144–146. https://doi.org/10.1007/s12262-008-0040-x

[65] Karp JG, Whitman L, Convit A (1991) Intentional ingestion of foreign objects by male prison inmates. Psychiatric Serv 42(5):533–535. https://doi.org/10.1176/ps.42.5.533

[66] Kerestes T, Smith J (2019) Paper or plastic? a foreign body ingestion leading to small bowel obstruction. a case report. ARC J Clin Case Rep 5(2). https://doi.org/10.20431/2455-9806.0502002

[67] Kluyver T, Ragan-Kelley B, Peres F, et al (2016) Jupyter Notebooks - a publishing format for reproducible computational workflows. Positioning and Power in Academic Publishing: Players, Agents and Agendas, IOS Press

[68] Kobiela J, Mittlener S, Gorycki T, et al (2015) Vast collection of foreign bodies in the stomach presenting as acute gastrointestinal bleeding in a patient with schizophrenia. Endosc 47(S 01):E356–E357. https://doi.org/10.1055/s-0034-1392611

[69] Kumar A, Jazieh AR (2001) Case report of sideroblastic anemia caused by ingestion of coins. American Journal of Hematology 66(2):126–129. https://doi.org/10.1002/1096-8652(200102)66:2⟨126::AID-AJH1029⟩3.0.CO;2-J

[70] Kumar R, Soni N, Bhardwaj S, et al (2019) Intentional foreign body ingestion. Internal and emergency medicine 14(8):1331–1333. https://doi.org/10.1007/s11739-019-02183-4

[71] Lamport L (1984) Latex. URL https://www.latex-project.org

[72] Lee JH, Kim HC, Yang DM, et al (2012) What is the role of plain radiography in patients with foreign bodies in the gastrointestinal tract? Clin Imaging 36(5):447–454. https://doi.org/10.1016/j.clinimag.2011.11.017

[73] Lee TH, Kang YW, Kim HJ, et al (2007) Foreign objects in korean prisoners. Korean J Intern Med 22(4):275–278. https://doi.org/10.3904/kjim.2007.22.4.275

[74] Lerche W (1911) The esophagoscope in removing sharp foreign bodies from the esophagus. JAMA LVI(9):634. https://doi.org/10.1001/jama.1911.02560090004002

[75] Li QP, Ge XX, Ji GZ, et al (2013) Endoscopic retrieval of 28 foreign bodies in a 100-year-old female after attempted suicide. World J Gastroenterol 19(25):4091–4093. https://doi.org/10.3748/wjg.v19.i25.4091

[76] Liu S, de Blacam C, Lim FY, et al (2005) Magnetic foreign body ingestions leading to duodenocolonic fistula. J Paediatr Gastroenterol and Nutr 41(5):670–672. https://doi.org/10.1097/01.mpg.0000177703.99786.c9

[77] Losanoff JE, Kjossev KT (1996) Gastrointestinal 'crosses'. a new shade from an old palette. Arch Surg 131(2):166–169. https://doi.org/10.1001/archsurg.1996.01430140056015

[78] Losanoff JE, Kjossev KT, Losanoff HE (1997) Oesophageal "cross"–a sinister foreign body. J Accid Emerg Med 14(1):54–55. https://doi.org/10.1136/emj.14.1.54

[79] Machin D, Campbell MJ, Walters SJK (2008) Medical statistics: a textbook for the health sciences, 4th edn. Wiley, Chichester

[80] Mesfin T, Tekalegn Y, Aman M, et al (2022) Ingestion of metallic materials found in the stomach and first part of the duodenum of a schizophrenic man: Case report. Int Med Case Rep J 15:681–684. https://doi.org/10.2147/IMCRJ.S386883

[81] Met Office (2010) Cartopy: a cartographic python library with a matplotlib interface. URL https://cartopy.readthedocs.io/stable/

[82] Microsoft Corporation (2025) Microsoft excel. URL https://www.microsoft.com/excel

[83] Microsoft Corporation (2025) Visual studio code. URL https://code.visualstudio.com/

[84] Misra S, Jain V, Ahmad F, et al (2013) Metallic sewing needle ingestion presenting as acute abdomen. Niger J Clin Prac 16(4):540–543. https://doi.org/10.4103/1119-3077.116879

[85] Moehlau FG (1895) Gastrostomy in the seventeenth century. Buffalo Med J 35(5):395–397

[86] Moola S, Munn Z, Tufanaru C, et al (2020) Chapter 7: Systematic reviews of etiology and risk. In: JBI Manual for Evidence Synthesis. JBI, p 263–268, https://doi.org/10.46658/JBIMES-20-08

[87] Naji H, Isacson D, Svensson JF, et al (2012) Bowel injuries caused by ingestion of multiple magnets in children: a growing hazard. Pediatr Surg Int 28(4):367–374. https://doi.org/10.1007/s00383-011-3026-x

[88] Nickerson A, Byrow Y, O'Donnell M, et al (2019) The association between visa insecurity and mental health, disability and social engagement in refugees living in australia. Eur J Psychotraumatol 10(1). https://doi.org/10.1080/20008198.2019.1688129

[89] Ohno Y, Yoneda A, Enjoji A, et al (2005) Gastroduodenal fistula caused by ingested magnets. Gastrointest Endosc 61(1):109–110. https://doi.org/10.1016/s0016-5107(04)02387-9

[90] Page MJ, McKenzie JE, Bossuyt PM, et al (2021) The prisma 2020 statement: an updated guideline for reporting systematic reviews. BMJ p n71. https://doi.org/10.1136/bmj.n71

[91] Palese C, Al-Kawas FH (2012) Repeat intentional foreign body ingestion: the importance of a multidisciplinary approach. Gastroenterol Hepatol 8(7):485–486. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC3533227/

[92] Palta R, Sahota A, Bemarki A, et al (2009) Foreign-body ingestion: characteristics and outcomes in a lower socioeconomic population with predominantly intentional ingestion. Gastrointest Endosc 69(3 Pt 1):426–433. https://doi.org/10.1016/j.gie.2008.05.072

[93] Pantazopoulos I, Mavrovounis G, Mermiri M, et al (2022) Intentional ingestion of batteries and razor blades by a prisoner: A true emergency? Int J Prisoner Health 18(3):316–322. https://doi.org/10.1108/IJPH-06-2021-0054

[94] Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: Machine learning in python. J Mach Learn Res 12(85):2825–2830. https://doi.org/10.5555/1953048.2078195

[95] Peixoto A, Pereira P, Macedo G (2017) Gastrointestinal: Voluntary padlock ingestion. J Gastroenterol Hepatol 32(12):1910. https://doi.org/10.1111/jgh.13828

[96] Poynter BA, Hunter JJ, Coverdale JH, et al (2011) Hard to swallow: a systematic review of deliberate foreign body ingestion. Gen Hosp Psych 33(5):518–524. https://doi.org/10.1016/j.genhosppsych.2011.06.011

[97] Puggioni R (2014) Speaking through the body: detention and bodily resistance in italy. Citizsh Stud 18(5):562–577. https://doi.org/10.1080/13621025.2014.923707

[98] Python Software Foundation (2025) Python. URL https://www.python.org/downloads/

[99] Qureshi NA, Cherian N, Ben-Hamida A, et al (2016) Endoscopic retrieval of an intentionally ingested mobile phone in an adult: First case report of its kind. Ann Clin Case Rep 1. URL https://www.researchgate.net/publication/332935488_Endoscopic_Retrieval_of_an_Intentionally_Ingested_Mobile_Phone_in_an_Adult_First_Case_Report_of_its_Kind

[100] Ricote GC, Torre LR, De Ayala VP, et al (1985) Fiberendoscopic removal of foreign bodies of the upper part of the gastrointestinal tract. Surg Gynecol Obstet 160(6):499–504. URL https://pubmed.ncbi.nlm.nih.gov/4002103/

[101] Riva CG, Toti FAT, Siboni S, et al (2018) Unusual foreign body impacted in the upper oesophagus: original technique for transoral extraction. BMJ Case Rep 2018:bcr–2018–225241. https://doi.org/10.1136/bcr-2018-225241

[102] Saint JH (1929) Surgery of the esophagus. Arch Surg 19(1):53–128. https://doi.org/10.1001/archsurg.1929.01150010056003

[103] Sakellaridis T, Potaris K, Mallios D, et al (2008) An unusual case of a swallowed thermometer perforated in the mediastinum. Ann Thorac Surg https://doi.org/10.1016/j.athoracsur.2007.07.027

[104] Salazar J, Pacheco N, Jaramillo P, et al (2020) Ingestion of razor blades, a rare event: A case report in a psychiatric patient. J Surg Case Rep 2020(5). https://doi.org/10.1093/JSCR/RJAA094

[105] Seabold S, Perktold J (2010) Statsmodels: Econometric and statistical modeling with python. In: Proceedings of the 9th Python in Science Conference, vol 57. SciPy, pp 92–96, https://doi.org/10.25080/Majora-92bf1922-011

[106] Sharma J, Riyaz S, Kilpatrick WJ (2022) Multi-disciplinary approach to managing deliberate foreign body ingestion on the medical floor. J Acad Consult Liaison Psychiatry https://doi.org/10.1016/j.jaclp.2022.10.025

[107] Shen L (2022) Forestplot. URL https://pypi.org/project/forestplot/

[108] Sobnach S, Castillo F, Blanco Vinent R, et al (2011) Penetrating cardiac injury following sewing needle ingestion. Heart Lung Circ 20(7):479–481. https://doi.org/10.1016/j.hlc.2011.01.006

[109] Sultan N, Attar H, Sembawa H, et al (2024) A plastic bezoar causing bowel obstruction: A case of table cover ingestion. Int J Surg Case Rep 117:109506. https://doi.org/10.1016/j.ijscr.2024.109506

[110] Sundvall M, Tidemalm DH, Titelman DE, et al (2015) Assessment and treatment of asylum seekers after a suicide attempt: a comparative study of people registered at mental health services in a swedish location. BMC Psychiatry 15(1):235. https://doi.org/10.1186/s12888-015-0613-8

[111] Tammana VS, Valluru N, Sanderson A (2012) All the wrong places: an unusual case of foreign body ingestion and inhalation. Case Rep Gastroenterol 6(3):778–783. https://doi.org/10.1159/000346287

[112] Tanrikulu Y, Sen Tanrikulu C, Karaman S, et al (2015) Ingestion of multiple magnets for suicide. Hong Kong J Emerg Med https://doi.org/10.1177/102490791502200107

[113] Tanriver-Ayder E, Faes C, van de Casteele T, et al (2021) Comparison of commonly used methods in random effects meta-analysis: application to preclinical data in drug discovery research. BMJ Open Sci 5(1):e100074. https://doi.org/10.1136/bmjos-2020-100074

[114] Tay ET, Weinberg G, Levin TL (2004) Ingested magnets: the force within. Pediatr Emerg Care 20(7):466–467. https://doi.org/10.1097/01.pec.0000134926.03030.a7

[115] te Wildt BT, Tettenborn C, Schneider U, et al (2010) Swallowing foreign bodies as an example of impulse control disorder in a patient with intellectual disabilities. Psychiatry 7(9):34–37

[116] Thapa N, Basnyat S, Maharjan M (2019) Ingestion of bell clappers by a shaman in jumla, nepal: A case report. J Nepal Med Assoc 57(215):56–58. https://doi.org/10.31729/jnma.4055

[117] The Pandas Development Team (2020) pandas-dev/pandas: Pandas. Zenodo, https://doi.org/10.5281/zenodo.13819579

[118] Trgo G, Tonkic A, Simunic M, et al (2012) Successful endoscopic removal of a lighter swallowed 17 months before. Case Rep Gastroenterol 6(2):238–242. https://doi.org/10.1159/000338839

[119] Tromans S, Chester V, Wells H, et al (2019) Deliberate ingestion of foreign bodies as a form of self-harm among inpatients within forensic mental health and intellectual disability services. J Forens Psychiatry Psychol 30(2):189–202. https://doi.org/10.1080/14789949.2018.1530287

[120] Tupesis JP, Kaminski A, Patel H, et al (2004) A penny for your thoughts: small bowel obstruction secondary to coin ingestion. J Emerg Med 27(3):249–252. https://doi.org/10.1016/j.jemermed.2004.03.013

[121] Udemgba C, Missov E, Percy R, et al (2021) A case report of an unusual left atrial mass. Eur Heart J Case Rep 5(1):ytaa500. https://doi.org/10.1093/ehjcr/ytaa500

[122] UN General Assembly (1967) Protocol relating to the status of refugees. URL https://www.ohchr.org/en/instruments-mechanisms/instruments/protocol-relating-status-refugees

[123] United Nations (1951) The 1951 refugee convention and 1967 protocol relating to the status of refugees. URL https://www.unhcr.org/media/convention-and-protocol-relating-status-refugees

[124] United Nations High Commissioner for Refugees (2022) A record 100 million people forcibly displaced worldwide. URL https://news.un.org/en/story/2022/05/1118772, accessed on 29 October 2024

[125] United Nations High Commissioner for Refugees (2025) Asylum-seekers. URL https://www.unhcr.org/us/about-unhcr/who-we-protect/asylum-seekers

[126] von Werthern M, Robjant K, Chui Z, et al (2018) The impact of immigration detention on mental health: a systematic review. BMC Psychiatry 18(1):382. https://doi.org/10.1186/s12888-018-1945-y

[127] Vujkovic M (2019) The suffering of omid: How a burns victim died despite injuries that were 'very survivable'. ABC News URL https://au.headtopics.com/news/the-suffering-of-omid-how-a-burns-victim-died-despite-injuries-that-were-very-survivable-4487685

[128] Wadhwa C, Madhavan S, Augustine AJ, et al (2015) The "mule" with golden eggs: Retrieval of unusual foreign body. J Dig Endosc https://doi.org/10.4103/0976-5042.159247

[129] Waskom ML (2021) seaborn: statistical data visualization. Journal of Open Source Software 6(60):3021. https://doi.org/10.21105/joss.03021

[130] Wildhaber BE, Le Coultre C, Genin B (2005) Ingestion of magnets: innocent in solitude, harmful in groups. J Pediatr Surg 40(10):e33–35. https://doi.org/10.1016/j.jpedsurg.2005.06.022

[131] Wnek B, Lozynska-Nelke A, Karon J (2015) Foreign body in the gastrointestinal tract leading to small bowel obstruction–case report and literature review. Pol Prz Chir 86(12):594–597. https://doi.org/10.1515/pjs-2015-0006

[132] Yasin MA, Malik GN, Malik SA, et al (2009) Metal in stomach: a rare cause of gastric bezoar. BMJ Case Rep 2009:bcr06.2008.0278. https://doi.org/10.1136/bcr.06.2008.0278

[133] Yıldız İ, Koca YS, Avşar G, et al (2016) Tendency to ingest foreign bodies in mentally retarded patients: A case with ileal perforation caused by the ingestion of a teaspoon. Case Rep Surg 2016:8075432. https://doi.org/10.1155/2016/8075432

[134] Zhao G, Zhao S, Wang S, et al (2022) Unexpected death from hepatic abscess 16 months after toothbrush ingestion. J Forens Sci https://doi.org/10.1111/1556-4029.15079